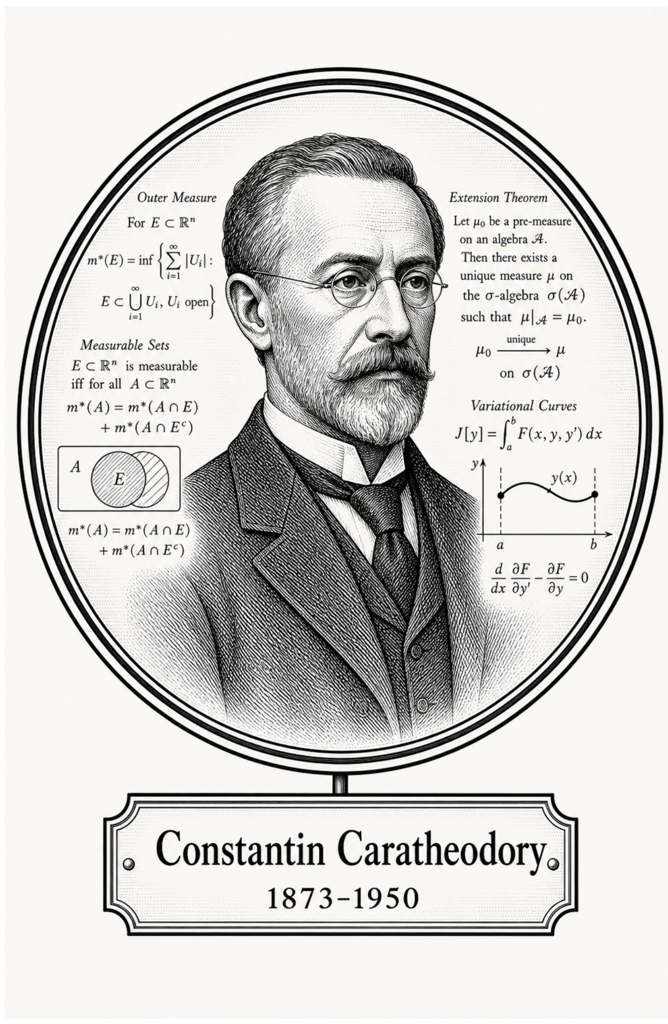
From Cantor to Ito

Mathematical Spaces



Constantin Caratheodory

1873-1950

Self-Study Notes

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For every reader learning to make mathematics their own.

Contents

Part I

Mathematical Spaces

Chapter 1

Mathematical Spaces

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1.1 Spaces as Structured Arenas

Definition — Space

Definition 1.1.1 (Space)**.** A **space** is a nonempty set.

Remark (Standard quantified statement)*.*

$$X \text{ is a space} \iff X \neq \varnothing.$$

Remark (Interpretation)*.* At the base level, the word "space" records that there is an arena of points under discussion. No distance, topology, operation, or measure has yet been chosen. Later phrases such as metric space, topological space, and measure space add structure to this underlying nonempty set.

Remark (Historical note)*.* This convention appears as Definition 2.1 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Exposition)*.* A mathematical space is an arena in which a particular kind of mathematics can be performed. At the base there is a set of objects. Added structure then specifies the ground rules: which objects can be compared, which subsets are distinguished, which operations are allowed, which limits make sense, or which quantities can be measured. The same underlying set can support many different spaces because different structures make different questions meaningful.

Remark (Structural Pattern)*.* The general pattern is

$$\text{space} = \text{underlying set} + \text{chosen structure}.$$

A metric space chooses a distance function. A topological space chooses a collection of open sets. A measurable space chooses a σ-algebra of measurable sets, and a measure space adds a measure to that measurable structure. Algebraic spaces choose operations and laws. Each choice controls which statements are legitimate inside that setting.

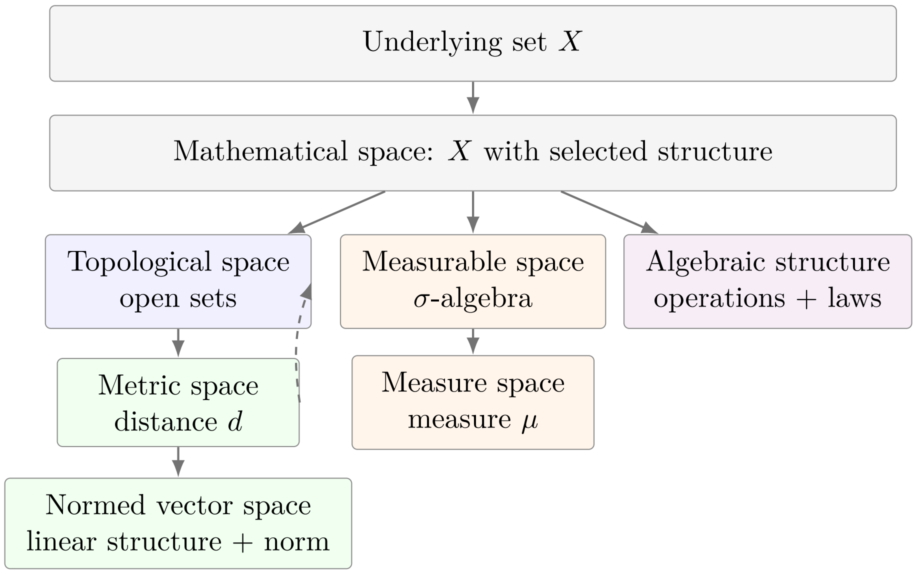
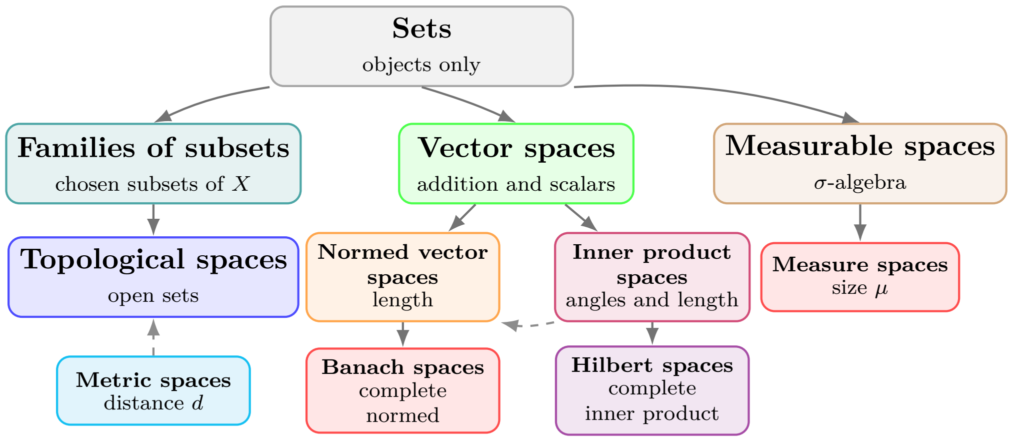


Figure 1.1: Mathematical spaces are built by adding structure to an underlying set. Some structures refine others: a metric induces a topology, while a measure requires measurable sets.

Remark (Interpretation). The nesting in Figure 1.1 should be read as an increase in available structure, not as a claim that every subject lies on a single chain. Metric spaces sit naturally over topology because every metric determines open balls and hence open sets. Measure spaces follow a different branch: they organize size rather than nearness. The common feature is the same: mathematical work begins only after the arena and its rules have been specified.

Remark (Exposition). The hierarchy in Figure 1.2 is a guide to structure, not a universal taxonomy. Some spaces are formed by adding data to a set. Others are formed by adding compatibility requirements to an already structured space. The following table records the construction pattern that will be used throughout this volume.

Remark (Structure Table).



The branches record added structure. A topology selects open sets, a metric supplies distance and induces open sets, linear spaces add operations, and measure spaces add measurable sets together with size.

Figure 1.2: One common route from sets to structured spaces. A topology begins with selected open subsets of a set. A metric gives a distance and thereby generates open balls, so metric spaces naturally carry topologies. Linear, normed, inner product, Banach, and Hilbert spaces add further compatible structure.

Space	**Comes from**	**Additional structure or rules**
Set	Primitive arena	A collection of objects; no nearness, size, or operation is specified.
Family of subsets	Set X	A selected collection $\mathcal{A} \subseteq \mathcal{P}(X)$.
Topological space	Set X	A topology $\mathcal{T}$: open sets containing $\varnothing$ and X, closed under arbitrary unions and finite intersections.
Metric space	Set X	A metric $d : X \times X \to [0, \infty)$ satisfying identity of indiscernibles, symmetry, and the triangle inequality; the metric induces open balls and hence a topology.
Vector space	Set V	Addition and scalar multiplication satisfying the vector space axioms over a chosen field.
Normed vector space	Vector space	A norm $\|\cdot\|$ measuring vector length, compatible with scalar multiplication and addition.
Inner product space	Vector space	An inner product $\langle \cdot, \cdot \rangle$ measuring angles and lengths; it induces a norm.
Banach space	Normed vector space	Completeness: every Cauchy sequence converges in the norm.
Hilbert space	Inner product space	Completeness with respect to the norm induced by the inner product.
Measurable space	Set X	A σ-algebra $\mathcal{M}$ of subsets regarded as measurable.
Measure space	Measurable space	A measure $\mu : \mathcal{M} \to [0, \infty]$ assigning size and satisfying countable additivity.

Remark (Structural Cards). The following cards use the structural presentation layer: each card records the inherited arena, the new data or laws, and the role of the resulting space. These are blueprint artifacts, not formal definitions; formal definitions remain authoritative when they are introduced.

Structural Card: Set

TAXONOMY

Primitive arena.

NEW AT

A nonempty domain X of objects.

AXIOMS / THEORY

No nearness, size, operation, or distinguished family of subsets is specified.

Structural Card: Family of Subsets

INHERITS

Set X.

NEW AT

A chosen collection $\mathcal{A} \subseteq \mathcal{P}(X)$.

TAXONOMY

Subset-selection structure. No closure law is imposed at this level.

Structural Card: Topological Space

INHERITS

Set X with a distinguished family of subsets.

NEW AT

A topology $\mathcal{T} \subseteq \mathcal{P}(X)$, read as the open subsets of X.

AXIOMS / THEORY

$\varnothing, X \in \mathcal{T}$; arbitrary unions of members of $\mathcal{T}$ lie in $\mathcal{T}$; finite intersections of members of $\mathcal{T}$ lie in $\mathcal{T}$.

TAXONOMY

Open-set structure for continuity, convergence, interior, closure, and neighborhood arguments.

Structural Card: Metric Space

INHERITS

Set X.

NEW AT

A distance function
$$d : X \times X \to [0, \infty).$$

AXIOMS / THEORY

Identity of indiscernibles, symmetry, and the triangle inequality.

USES-A

Open balls $B(x, r) = \{y \in X : d(x, y) < r\}$.

TAXONOMY

Distance structure. The metric induces a topology by declaring sets open when they contain a metric ball around each of their points.

Structural Card: Vector Space

INHERITS

Set V and a scalar field $\mathbb{F}$.

NEW AT

Vector addition $+ : V \times V \to V$, scalar multiplication $\cdot : \mathbb{F} \times V \to V$, and a zero vector $0 \in V$.

AXIOMS / THEORY

$(V, +, 0)$ is an abelian group; scalar multiplication is unital and associative; scalar multiplication distributes over vector addition and field addition.

TAXONOMY

Linear structure. The space supports linear combinations before any length, angle, or limit structure is chosen.

Structural Card: Normed Vector Space

INHERITS

Vector space V over $\mathbb{F}$.

NEW AT

A norm $\|\cdot\| : V \to [0, \infty)$.

AXIOMS / THEORY

$\|v\| = 0$ if and only if $v = 0$; $\|\alpha v\| = |\alpha| \, \|v\|$; $\|u + v\| \leq \|u\| + \|v\|$.

USES-A

The induced metric $d(u, v) = \|u - v\|$.

TAXONOMY

Linear metric structure. Algebraic operations and metric notions become compatible.

Structural Card: Inner Product Space

INHERITS

Real or complex vector space V.

NEW AT

An inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$.

AXIOMS / THEORY

Positive definiteness, conjugate symmetry, and linearity in the chosen argument.

USES-A

The induced norm $\|v\| = \sqrt{\langle v, v \rangle}$.

TAXONOMY

Linear geometry. Length, angle, and orthogonality are available, and the metric structure is derived from the inner product.

Structural Card: Banach Space

INHERITS

Normed vector space $(V, \|\cdot\|)$.

NEW AT

Completeness with respect to the norm-induced metric.

AXIOMS / THEORY

Every Cauchy sequence in V converges to a point of V in the norm.

TAXONOMY

Complete normed linear space. Approximation by Cauchy sequences remains inside the space.

Structural Card: Hilbert Space

INHERITS

Inner product space H.

NEW AT

Completeness with respect to the norm induced by the inner product.

AXIOMS / THEORY

Every Cauchy sequence in the induced norm converges to a point of H.

TAXONOMY

Complete inner product space. Orthogonality and projection arguments take place in a complete metric environment.

Structural Card: Measurable Space

INHERITS

Set X with a distinguished family of subsets.

NEW AT

A σ-algebra $\mathcal{M} \subseteq \mathcal{P}(X)$, read as the measurable subsets of X.

AXIOMS / THEORY

$X \in \mathcal{M}$; complements of measurable sets are measurable; countable unions of measurable sets are measurable.

TAXONOMY

Countably closed subset-selection structure. It specifies which subsets are legal inputs for size assignments.

Structural Card: Measure Space

INHERITS

Measurable space $(X, \mathcal{M})$.

NEW AT

A measure $\mu : \mathcal{M} \to [0, \infty]$.

AXIOMS / THEORY

$\mu(\varnothing) = 0$, and μ is countably additive on pairwise disjoint measurable sets:

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

TAXONOMY

Quantitative measurable structure. The σ-algebra determines the admissible subsets, and the measure assigns their size.

Proofs

Chapter 2

Metric Spaces

SERIES
From Cantor to Ito

VOLUME IV
Mathematical Spaces

BOOK I
Mathematical Spaces

CHAPTER 2
Metric Spaces

2.1 Metric Spaces

2.2 Definitions

Definition 2.2.1 (Metric Space)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}$$

is called a *metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$ *(positive definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

A *metric space* is a pair (X, d), where X is a nonempty set and d is a metric on X.

Remark (Standard quantified statement)*.*

$$\begin{aligned}\mathrm{Metric}(X, d) \iff X \neq \varnothing \ \wedge \ & d : X \times X \to \mathbb{R} \\ & \wedge \forall x, y, z \in X \colon d(x, y) \geq 0 \wedge \big(d(x, y) = 0 \iff x = y\big) \\ & \wedge d(x, y) = d(y, x) \wedge d(x, y) \leq d(x, z) + d(z, y).\end{aligned}$$

Remark (Interpretation)*.* A metric turns a set into a distance arena. Positive definiteness says that distance is never negative and that zero distance identifies exactly one point. Symmetry says that distance does not depend on direction. The triangle inequality says that moving through an intermediate point cannot make the direct distance larger than the two-step route.

Remark (Historical note). This convention appears as Definition 2.2 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Definition 2.2.2 (Pseudo-Metric)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}$$

is called a *pseudo-metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, x) = 0$ *(positive semi-definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

Remark (Standard quantified statement).

$$\begin{aligned}\operatorname{PseudoMetric}(X, d) \iff & X \neq \varnothing \ \wedge \ d : X \times X \to \mathbb{R} \\ & \wedge \forall x, y, z \in X\colon d(x, y) \geq 0 \wedge d(x, x) = 0 \\ & \wedge d(x, y) = d(y, x) \wedge d(x, y) \leq d(x, z) + d(z, y).\end{aligned}$$

Remark (Interpretation). A pseudo-metric keeps the distance-like laws of nonnegativity, symmetry, and the triangle inequality, but it weakens positive definiteness. Distinct points may have distance 0. Thus a pseudo-metric measures separation only up to possible zero-distance identifications.

Remark (Historical note). This convention appears as Definition 2.6 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Lemma

Lemma 2.2.3 (Metrics are Pseudo-Metrics)**.** *Every metric on a nonempty set X is a pseudo-metric on X. The converse is false: there exist pseudo-metrics that are not metrics.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall X\, \forall d\colon\ \operatorname{Metric}(X, d) \implies \operatorname{PseudoMetric}(X, d), \\ &\exists X\, \exists d\colon\ \operatorname{PseudoMetric}(X, d) \wedge \neg \operatorname{Metric}(X, d).\end{aligned}$$

Remark (Interpretation). The implication is immediate because the metric axioms contain all pseudo-metric axioms. The only difference is the zero-distance law: a metric forces $d(x, y) = 0$ exactly when $x = y$, while a pseudo-metric only requires $d(x, x) = 0$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Pseudo-Metric

2.3 Metrics

Remark (Exposition). A metric is the distance function itself. A metric space is the set together with that function. This section records standard metric functions and then states, one by one, that each function satisfies the metric axioms.

2.3.1 Norms Induce Metrics

Definition — Norm

Definition 2.3.1 (Norm)**.** Let V be a real vector space. A *norm* on V is a function

$$\|\cdot\| : V \to \mathbb{R}_{\geq 0}$$

such that for all $x, y \in V$ and all $\lambda \in \mathbb{R}$,

1. $\|x\| = 0$ if and only if $x = 0$ *(positive definiteness)*;
2. $\|\lambda x\| = |\lambda| \, \|x\|$ *(absolute homogeneity)*;
3. $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality)*.

Remark (Standard quantified statement).

$$\begin{aligned}\mathrm{Norm}(V, \|\cdot\|) \iff{} & \|\cdot\| : V \to \mathbb{R}_{\geq 0} \\ & \wedge \forall x \in V \colon \|x\| = 0 \iff x = 0 \\ & \wedge \forall \lambda \in \mathbb{R} \; \forall x \in V \colon \|\lambda x\| = |\lambda| \, \|x\| \\ & \wedge \forall x, y \in V \colon \|x + y\| \leq \|x\| + \|y\|.\end{aligned}$$

Remark (Interpretation). A norm measures the size of one vector. The metric induced by a norm measures the size of the displacement vector $x - y$. Thus a norm is a length structure, and the induced metric is the corresponding distance structure.

Theorem

Theorem 2.3.2 (A Norm Induces a Metric)**.** *Let V be a real vector space and let $\|\cdot\|$ be a norm on V. Define*

$$d_{\|\cdot\|} : V \times V \to \mathbb{R}_{\geq 0}, \qquad d_{\|\cdot\|}(x, y) := \|x - y\|.$$

Then $d_{\|\cdot\|}$ is a metric on V.
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Norm}(V, \|\cdot\|) \Longrightarrow \mathrm{Metric}(V, d_{\|\cdot\|}), \qquad d_{\|\cdot\|}(x, y) = \|x - y\|.$$

Remark (Interpretation). The metric axioms are exactly the norm axioms applied to displacement vectors. Zero distance says the displacement $x - y$ has zero norm, hence is the zero vector. Symmetry follows from absolute homogeneity with scalar -1. The triangle inequality follows by writing $x - y = (x - z) + (z - y)$.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Norm

Definition — Discrete Metric

Definition 2.3.3 (Discrete Metric)**.** Let X be a nonempty set. The *discrete metric* on X is the function

$$d_{\text{disc}} : X \times X \to \mathbb{R}_{\geq 0}$$

defined by

$$d_{\text{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Remark (Standard quantified statement).

$$\begin{aligned} &X \neq \varnothing, && d_{\text{disc}} : X \times X \to \mathbb{R}_{\geq 0}, \\ &\forall x, y \in X\colon && d_{\text{disc}}(x, y) = 0 \iff x = y, \\ &\forall x, y \in X\colon && x \neq y \iff d_{\text{disc}}(x, y) = 1. \end{aligned}$$

Remark (Interpretation). The discrete metric treats equality as the only closeness relation. Distinct points are all placed at the same positive distance from each other.

Proposition

Proposition 2.3.4 (The Discrete Distance is a Metric)**.** *For every nonempty set X, the discrete distance d_{disc} is a metric on X.*
Go to proof.

Remark (Standard quantified statement).

$$X \neq \varnothing \implies \mathrm{Metric}(X, d_{\text{disc}}).$$

Remark (Interpretation). The only point at distance 0 from x is x itself. Symmetry is built into the two-case definition, and the triangle inequality is automatic because the only possible distances are 0 and 1.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Discrete Metric

Definition — Euclidean Distance on the Real Line

Definition 2.3.5 (Euclidean Distance on $\mathbb{R}$)**.** The *Euclidean distance* on $\mathbb{R}$ is the function

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\geq 0}, \qquad d_{\mathbb{R}}(x, y) = |x - y|.$$

Remark (Standard quantified statement).

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\geq 0}, \qquad \forall x, y \in \mathbb{R}: \quad d_{\mathbb{R}}(x, y) = |x - y|.$$

Remark (Interpretation). The real-line distance records absolute displacement between two real numbers. It forgets direction and keeps only separation.

Proposition

Proposition 2.3.6 (The Real-Line Euclidean Distance is a Metric)**.** *The function $d_{\mathbb{R}}(x, y) = |x - y|$ is a metric on $\mathbb{R}$.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}, d_{\mathbb{R}}).$$

Remark (Interpretation). This proposition says that the usual absolute-value distance satisfies the abstract metric axioms.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}$

Definition — Euclidean Space

Definition 2.3.7 (Euclidean Space)**.** For $n \in \mathbb{N}$, the *n-dimensional Euclidean space* is $\mathbb{R}^n$ equipped with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

Remark (Standard quantified statement). For each $n \in \mathbb{N}$, $\mathrm{EuclideanSpace}(n)$ denotes $\mathbb{R}^n$ together with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

Remark (Interpretation). Euclidean space is the coordinate model of ordinary distance geometry.

Remark (Historical note). This convention is the setting for the Euclidean examples in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Definition — Euclidean Distance on $\mathbb{R}^n$

Definition 2.3.8 (Euclidean Distance on $\mathbb{R}^n$)**.** For $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$, the *Euclidean distance* is the function

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_2(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

Remark (Standard quantified statement).

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_2(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

Remark (Interpretation). The Euclidean distance measures the length of the coordinate difference vector $x - y$.

Theorem

Theorem 2.3.9 (Cauchy–Schwarz Inequality)**.** *For every $x, y \in \mathbb{R}^n$,*

$$|x \cdot y| \leq |x|\,|y|.$$

Equality holds if and only if x and y are linearly dependent.
Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}^n\colon \quad |x \cdot y| \leq |x|\,|y|.$$

Remark (Interpretation). Cauchy–Schwarz controls the dot product by the lengths of the two vectors. It is the algebraic estimate behind Euclidean triangle inequalities.

Remark (Dependencies).

- Definition: Euclidean Space
- Definition: Euclidean Distance on $\mathbb{R}^n$

Corollary

Corollary 2.3.10 (Minkowski's Inequality)**.** *For every $x, y \in \mathbb{R}^n$,*

$$|x + y| \leq |x| + |y|.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}^n\colon \quad |x + y| \leq |x| + |y|.$$

Remark (Interpretation). Minkowski's inequality is the triangle inequality for Euclidean length.

Remark (Dependencies).

- Theorem: Cauchy–Schwarz Inequality
- Definition: Euclidean Distance on $\mathbb{R}^n$

Proposition

Proposition 2.3.11 (The Euclidean Distance on $\mathbb{R}^n$ is a Metric)**.** *The function d_2 is a metric on $\mathbb{R}^n$.*

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{Metric}(\mathbb{R}^n, d_2).$$

Remark (Interpretation)*.* The Euclidean distance makes $\mathbb{R}^n$ into the standard finite dimensional metric space.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Minkowski's Inequality

Corollary

Corollary 2.3.12 (Euclidean Metric)**.** *For every $x, y \in \mathbb{R}^n$, define*

$$d_2(x, y) := \sqrt{|x_1 - y_1|^2 + \cdots + |x_n - y_n|^2}.$$

Then $(\mathbb{R}^n, d_2)$ is a metric space.

Go to proof.

Remark (Standard quantified statement)*.*

$$\text{Metric}(\mathbb{R}^n, d_2).$$

Remark (Interpretation)*.* This records the standard Euclidean metric-space structure on $\mathbb{R}^n$.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Minkowski's Inequality

Definition — Taxicab Metric

Definition 2.3.13 (Taxicab Metric)**.** For $x, y \in \mathbb{R}^n$, the *taxicab metric* is the function

$$d_1(x, y) := \sum_{i=1}^{n} |x_i - y_i|.$$

Remark (Standard quantified statement).

$$d_1 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}, \qquad d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i|.$$

Remark (Interpretation). Taxicab distance measures coordinate displacement additively, as if travel were constrained to coordinate-parallel streets.

Proposition

Proposition 2.3.14 (Taxicab Metric). *The function d_1 is a metric on $\mathbb{R}^n$.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{R}^n, d_1).$$

Remark (Interpretation). The taxicab metric satisfies the metric axioms coordinate by coordinate, with the real absolute-value triangle inequality summed over all coordinates.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Taxicab Metric

Definition — p-Metric on $\mathbb{R}^n$

Definition 2.3.15 (p-Metric on $\mathbb{R}^n$). Let $1 \leq p < \infty$. For $x, y \in \mathbb{R}^n$, define

$$d_p(x, y) := \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}.$$

Remark (Standard quantified statement).

$$1 \leq p < \infty \implies d_p : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\geq 0}.$$

Remark (Interpretation). The p-metrics interpolate between taxicab distance at $p = 1$ and Euclidean distance at $p = 2$, and they form the standard finite-dimensional ℓ^p family.

Theorem

Theorem 2.3.16 (d_p is a Metric). *For every $1 \leq p < \infty$, the function d_p is a metric on $\mathbb{R}^n$.*
Go to proof.

Remark (Standard quantified statement).

$$1 \leq p < \infty \implies \mathrm{Metric}(\mathbb{R}^n, d_p).$$

Remark (Interpretation). The proof is the p-norm version of Minkowski's inequality. The theorem packages a whole family of metric structures on the same underlying set.

Remark (Dependencies).

- Definition: Metric Space
- Definition: p-Metric on $\mathbb{R}^n$

Definition — Complex Metric

Definition 2.3.17 (Complex Metric)**.** Identify $\mathbb{C}$ with $\mathbb{R}^2$. The *complex metric* is

$$d_{\mathbb{C}}(z, w) := |z - w|.$$

Remark (Standard quantified statement).

$$d_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{R}_{\geq 0}, \qquad d_{\mathbb{C}}(z, w) = |z - w|.$$

Remark (Interpretation). As a distance function, the complex metric is the Euclidean metric on $\mathbb{R}^2$ written in complex notation.

Proposition

Proposition 2.3.18 (The Complex Plane is a Metric Space)**.** *The pair* $(\mathbb{C}, d_{\mathbb{C}})$ *is a metric space.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\mathbb{C}, d_{\mathbb{C}}).$$

Remark (Interpretation). The metric on $\mathbb{C}$ is inherited from the Euclidean plane.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Complex Metric
- Corollary: Euclidean Metric

Definition — Supremum Metric on Bounded Real Sequences

Definition 2.3.19 (Supremum Metric on Bounded Real Sequences)**.** Let $\ell^\infty(\mathbb{R})$ denote the set of all bounded real sequences:

$$\ell^\infty(\mathbb{R}) := \{x = (x_n)_{n \in \mathbb{N}} : \exists M \geq 0 \; \forall n \in \mathbb{N} \; |x_n| \leq M\}.$$

For $x, y \in \ell^\infty(\mathbb{R})$, define

$$d_\infty(x, y) := \sup_{n \in \mathbb{N}} |x_n - y_n|.$$

Remark (Standard quantified statement).

$$d_\infty : \ell^\infty(\mathbb{R}) \times \ell^\infty(\mathbb{R}) \to \mathbb{R}_{\geq 0}, \qquad d_\infty(x, y) = \sup_{n \in \mathbb{N}} |x_n - y_n|.$$

Remark (Interpretation). The supremum metric measures the largest coordinatewise discrepancy between two bounded sequences.

Proposition

Proposition 2.3.20 (The Supremum Distance on Bounded Sequences is a Metric)**.** *The function d_∞ is a metric on $\ell^\infty(\mathbb{R})$.*
Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Metric}(\ell^\infty(\mathbb{R}), d_\infty).$$

Remark (Interpretation). Two bounded sequences are close in this metric when every coordinate is close by the same tolerance.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Supremum Metric on Bounded Real Sequences

2.4 Examples of Metric Spaces

Remark (Exposition). The preceding metrics become metric spaces once their domains are paired with the verified distance functions. This section records the resulting spaces as objects in their own right.

Definition — Discrete Metric Space

Definition 2.4.1 (Discrete Metric Space)**.** Let X be a nonempty set. The *discrete metric space on X* is the metric space

$$(X, d_{\mathrm{disc}}),$$

where d_{disc} is the discrete metric on X.

Remark (Standard quantified statement).

$$X \neq \varnothing \implies \mathrm{MetricSpace}(X, d_{\mathrm{disc}}).$$

Remark (Interpretation). A discrete metric space is a space where distinct points are all separated by the same positive distance. Its geometry records equality and inequality, but no finer notion of nearness.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Discrete Metric
- Proposition: The Discrete Distance is a Metric

Definition — Euclidean Metric Space

Definition 2.4.2 (Euclidean Metric Space)**.** For $n \in \mathbb{N}$, the *standard Euclidean metric space* is

$$(\mathbb{R}^n, d_2),$$

where d_2 is the Euclidean metric on $\mathbb{R}^n$.

Remark (Standard quantified statement).

$$n \in \mathbb{N} \Longrightarrow \mathrm{MetricSpace}(\mathbb{R}^n, d_2).$$

Remark (Interpretation). The Euclidean metric space is the standard finite-dimensional distance model: points are coordinate vectors, and distance is ordinary straight-line length.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Definition — Taxicab Metric Space

Definition 2.4.3 (Taxicab Metric Space)**.** For $n \in \mathbb{N}$, the *taxicab metric space* is

$$(\mathbb{R}^n, d_1),$$

where d_1 is the taxicab metric on $\mathbb{R}^n$.

Remark (Standard quantified statement).

$$n \in \mathbb{N} \Longrightarrow \mathrm{MetricSpace}(\mathbb{R}^n, d_1).$$

Remark (Interpretation). The taxicab metric space has the same underlying set as Euclidean space, but distance is measured by summing coordinatewise displacements.

Remark (Dependencies).

- Definition: Metric Space

- Definition: Taxicab Metric
- Proposition: Taxicab Metric

Definition — p-Metric Space

Definition 2.4.4 (p-Metric Space)**.** Let $1 \leq p < \infty$ and let $n \in \mathbb{N}$. The *p-metric space* is

$$(\mathbb{R}^n, d_p),$$

where d_p is the p-metric on $\mathbb{R}^n$.

Remark (Standard quantified statement)*.*

$$1 \leq p < \infty,\ n \in \mathbb{N} \Longrightarrow \mathrm{MetricSpace}(\mathbb{R}^n, d_p).$$

Remark (Interpretation)*.* The p-metric spaces form a family of finite-dimensional metric spaces on the same coordinate set, with different rules for turning coordinate displacements into distance.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: p-Metric on $\mathbb{R}^n$
- Theorem: d_p is a Metric

Definition — Complex Metric Space

Definition 2.4.5 (Complex Metric Space)**.** The *complex metric space* is

$$(\mathbb{C}, d_{\mathbb{C}}),$$

where $d_{\mathbb{C}}$ is the usual Euclidean distance on the complex plane.

Remark (Standard quantified statement)*.*

$$\mathrm{MetricSpace}(\mathbb{C}, d_{\mathbb{C}}).$$

Remark (Interpretation)*.* The complex metric space treats complex numbers as points in the Euclidean plane, while retaining the notation and algebra of $\mathbb{C}$.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Complex Metric
- Proposition: The Complex Plane is a Metric Space

Definition — Bounded Sequence Metric Space

Definition 2.4.6 (Bounded Sequence Metric Space)**.** The *bounded sequence metric space* is

$$(\ell^\infty(\mathbb{R}), d_\infty),$$

where $\ell^\infty(\mathbb{R})$ is the set of bounded real sequences and d_∞ is the supremum metric.

Remark (Standard quantified statement)*.*

$$\mathrm{MetricSpace}(\ell^\infty(\mathbb{R}), d_\infty).$$

Remark (Interpretation)*.* The bounded sequence metric space is an infinite-dimensional example. Its distance measures the largest coordinatewise discrepancy between two bounded real sequences.

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Supremum Metric on Bounded Real Sequences
- Proposition: Supremum Metric on Bounded Sequences

2.5 Balls and Bounded Sets

Definition — Open Ball

Definition 2.5.1 (Open Ball)**.** Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *open ball* of radius r centered at x is

$$B_d(x; r) := \{\, y \in X : d(y, x) < r \,\}.$$

Remark (Standard quantified statement)*.*

$$\forall y \in X: \quad y \in B_d(x; r) \iff d(y, x) < r.$$

Remark (Interpretation)*.* An open ball collects exactly the points whose distance from the center is strictly less than the chosen radius. The strict inequality leaves the boundary points out.

Remark (Historical note)*.* This notation follows Definition 2.15 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies)*.*

- Definition: Metric Space

Definition — Closed Ball

Definition 2.5.2 (Closed Ball)**.** Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *closed ball* of radius r centered at x is

$$\overline{B}_d(x; r) := \{\, y \in X : d(y, x) \le r \,\}.$$

Remark (Standard quantified statement).

$$\forall y \in X\colon \quad y \in \overline{B}_d(x; r) \iff d(y, x) \le r.$$

Remark (Interpretation). A closed ball uses the same center and radius as the corresponding open ball, but it includes the boundary points at distance exactly r.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

2.6 Open and Closed Sets

Remark (Exposition). Open and closed sets describe how subsets sit inside a metric space. Open sets are subsets that contain a small ball around each of their points. Closed sets are subsets whose complements are open.

Definition — Metric Interior Point

Definition 2.6.1 (Metric Interior Point)**.** Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. The point x is an *interior point of* E if there exists $\delta > 0$ such that

$$B_d(x; \delta) \subseteq E.$$

Remark (Standard quantified statement).

$$x \text{ is an interior point of } E \quad \iff \quad x \in X \ \wedge \ \exists \delta > 0\colon B_d(x; \delta) \subseteq E.$$

Remark (Interpretation). An interior point of E is a point with positive-radius room inside E. The entire ball around the point must remain in E, not merely the point itself.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Definition — Metric Interior

Definition 2.6.2 (Metric Interior)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *interior of* E is the set of all interior points of E. It is denoted by

$$E^\circ.$$

Remark (Standard quantified statement).

$$E^\circ = \{x \in X : \exists \delta > 0 \text{ such that } B_d(x;\delta) \subseteq E\}.$$

Remark (Interpretation). The interior E° collects exactly the points of E that have positive-radius room inside E.

Remark (Notation). The notation E° is read as the *interior of* E or E*-interior*.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Interior Point

Definition — Metric Open Set

Definition 2.6.3 (Metric Open Set)**.** Let (X,d) be a metric space. A subset $E \subseteq X$ is *open in* (X,d) if either $E = \varnothing$ or every $x \in E$ is an interior point of E.

Remark (Standard quantified statement).

$$E \text{ is open in } (X,d) \quad \Longleftrightarrow \quad E \subseteq X \ \wedge \ \big(E = \varnothing \ \vee \ \forall x \in E\, \exists \varepsilon > 0\colon B_d(x;\varepsilon) \subseteq E\big).$$

Remark (Interpretation). An open set contains room around each of its points. The amount of room may depend on the point, but every point of the set must have some positive-radius ball that stays inside the set.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Epsilon Neighborhood
- Definition: Metric Interior Point

Theorem

Theorem 2.6.4 (Metric Open Set Closure Properties)**.** *Let X be a metric space. Then:*

1. *the empty set $\varnothing$ and the space X are open sets;*
2. *an arbitrary union of open subsets of X is open;*
3. *any finite intersection of open subsets of X is open.*

Go to proof.

Remark (Standard quantified statement). If X is a metric space, then

$$\varnothing \text{ is open in } X, \qquad X \text{ is open in } X,$$

$$\{U_\alpha\}_{\alpha \in A} \text{ open in } X \Longrightarrow \bigcup_{\alpha \in A} U_\alpha \text{ is open in } X,$$

and, for $n \in \mathbb{N}$,

$$U_1, \ldots, U_n \text{ open in } X \Longrightarrow \bigcap_{k=1}^{n} U_k \text{ is open in } X.$$

Remark (Interpretation). Metric open sets are stable under the same basic operations that define a topology: arbitrary unions and finite intersections. Thus every metric determines a topology by declaring its metric-open subsets to be open.

Remark (Historical note). This result corresponds to Theorem 3.2 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Open Set
- Definition: Metric Interior Point

Theorem

Theorem 2.6.5 (Open Balls are Open). *Open balls in metric spaces are open sets.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (X, d)\, \forall x \in X\, \forall r > 0\colon \quad B_d(x; r) \text{ is open in } (X, d).$$

Remark (Interpretation). Every metric ball contains room around each of its points. The triangle inequality is the mechanism that transfers room around the center to room around an arbitrary point inside the ball.

Remark (Historical note). This result corresponds to Theorem 3.4 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Metric Open Set

Theorem

Theorem 2.6.6 (Interior is the Largest Open Subset)**.** *Let E be a subset of a metric space X. Then E° is the largest open subset of X contained in E. In other words:*

1. $E^\circ \subseteq E$;
2. E° *is open;*
3. $E^\circ \supseteq O$, *for every open set* $O \subseteq E$.

Go to proof.

Remark (Standard quantified statement).

$$E^\circ \subseteq E, \qquad E^\circ \text{ is open in } X, \qquad \forall O \subseteq E\colon O \text{ open in } X \implies O \subseteq E^\circ.$$

Remark (Interpretation). The interior operation extracts the open part of a set. It keeps exactly the points with room inside E, and it contains every open subset already lying inside E.

Remark (Historical note). This result corresponds to Theorem 3.5 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Interior
- Definition: Metric Open Set
- Theorem: Open Balls are Open

Definition — Metric Closed Set

Definition 2.6.7 (Metric Closed Set)**.** Let (X, d) be a metric space. A subset $E \subseteq X$ is *closed in* (X, d) if its complement $E^c = X \setminus E$ is open in (X, d).

Remark (Standard quantified statement).

$$E \text{ is closed in } (X,d) \quad \Longleftrightarrow \quad E \subseteq X \ \wedge \ E^c = X \setminus E \ \wedge \ E^c \text{ is open in } (X,d).$$

Remark (Interpretation). A closed set is defined by the openness of what lies outside it. In metric spaces this means every point outside E has some positive-radius ball that misses E.

Remark (Notation). When E is a subset of a fixed ambient space X, the symbol E^c denotes the complement of E in X:

$$E^c = X \setminus E.$$

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Open Set

2.7 Limit Points and Isolated Points

Remark (Exposition). Limit points and isolated points separate two ways a subset can contain or approach a point. A limit point is repeatedly approached by other points of the set at every scale. An isolated point belongs to the set but has a small ball in which it is the only point of the set.

Definition — Metric Limit Point and Isolated Point

Definition 2.7.1 (Metric Limit Point and Isolated Point)**.** Let E be a subset of a metric space (X, d), and let $x \in X$.

1. The point x is a *limit point of* E if, for every $\varepsilon > 0$, the ball $B_d(x; \varepsilon)$ contains a point of $E \setminus \{x\}$.

2. The point x is an *isolated point of* E if $x \in E$ and x is not a limit point of E.

Remark (Standard quantified statement).

$$x \text{ is a limit point of } E \iff \forall \varepsilon > 0\colon B_d(x; \varepsilon) \cap (E \setminus \{x\}) \neq \varnothing.$$

$$x \text{ is an isolated point of } E \iff x \in E \ \wedge \ \exists \varepsilon > 0\colon B_d(x; \varepsilon) \cap E = \{x\}.$$

Remark (Interpretation). A limit point of E may lie in E or outside E; what matters is that points of E distinct from x occur arbitrarily close to x. An isolated point must belong to E, and some positive-radius ball around it contains no other point of E.

Remark (Notation). The set of limit points of E is denoted by E', read as E*-prime*.

Remark (Exposition). Limit points are also called *cluster points* or *accumulation points*. In later work with functions on subsets of metric spaces, limits of the form $\lim_{y \to x} f(y)$ are only meaningful when x is a limit point of the domain being approached.

Remark (Examples). In the usual metric on $\mathbb{R}$, if

$$E = (0, 1) \cup \{2\},$$

then 2 is an isolated point of E, while $[0, 1]$ is the set of limit points of E.

There are also metric spaces with infinite subsets that have no limit points. For example, if X is an infinite discrete metric space and $E = X$, then $E' = \varnothing$. Also, in $X = \mathbb{R} \setminus \mathbb{Q}$, the set

$$E = \{\sqrt{2}/n : n \in \mathbb{N}\}$$

has no limit point in X.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Theorem

Theorem 2.7.2 (Closed Sets Contain Their Limit Points)**.** *Let X be a metric space and let $E \subseteq X$. Then E is closed if and only if $E' \subseteq E$.*

Go to proof.

Remark (Standard quantified statement).

$$E \text{ is closed in } X \iff E' \subseteq E.$$

Remark (Interpretation). Closed sets contain all points that can be approached by other points of the set. Equivalently, no limit point of a closed set is missing from the set.

Remark (Historical note). This result corresponds to Theorem 3.10 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Closed Set
- Definition: Metric Limit Point and Isolated Point

Theorem

Theorem 2.7.3 (Sequential Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if there exists a sequence of distinct terms in E that converges to x.*

Go to proof.

Remark (Standard quantified statement).

$$x \in E' \iff \exists \{x_n\} \subseteq E\colon \big(\forall n \neq m,\ x_n \neq x_m\big) \ \wedge\ x_n \to x.$$

Remark (Interpretation). Limit points can be detected sequentially. A point is a limit point exactly when the set supplies infinitely many distinct approximating terms converging to that point.

Remark (Historical note). This result corresponds to Theorem 3.11 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Limit Point and Isolated Point
- Definition: Metric Sequential Convergence

Corollary

Corollary 2.7.4 (Neighborhood Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if every neighborhood of x contains infinitely many points of E.*

Go to proof.

Remark (Standard quantified statement).

$$x \in E' \iff \forall U\colon \big(U \text{ is a neighborhood of } x\big) \implies U \cap E \text{ is infinite.}$$

Remark (Interpretation). The neighborhood form strengthens the ball definition into a count statement: near a limit point, every neighborhood contains not just one nearby point of E, but infinitely many.

Remark (Historical note). This result corresponds to Corollary 3.12 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Neighborhood
- Theorem: Sequential Characterization of Limit Points

Theorem

Theorem 2.7.5 (Bolzano–Weierstrass for Bounded Infinite Subsets). *Every bounded infinite subset of $\mathbb{R}^m$ has a limit point in $\mathbb{R}^m$.*

Go to proof.

Remark (Standard quantified statement).

$$E \subseteq \mathbb{R}^m, \quad E \text{ bounded}, \quad E \text{ infinite} \quad \implies \quad \exists x \in \mathbb{R}^m\colon x \in E'.$$

Remark (Interpretation). In finite-dimensional Euclidean space, boundedness prevents an infinite set from spreading out completely. Some point of the ambient space must be approached by infinitely many points of the set.

Remark (Historical note). This result corresponds to Theorem 3.13 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Theorem: Bolzano–Weierstrass in $\mathbb{R}^m$
- Theorem: Sequential Characterization of Limit Points

Proposition

Proposition 2.7.6 (The Derived Set is Closed). *The set of limit points of any set is closed.*

Go to proof.

Remark (Standard quantified statement).

$$E \subseteq X \quad \implies \quad E' \text{ is closed in } X.$$

Remark (Interpretation). Taking limit points produces a closed set. Once a point is approached by limit points of E, it is itself forced to be a limit point of E.

Remark (Historical note). This result corresponds to Proposition 3.15 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Limit Point and Isolated Point
- Theorem: Closed Sets Contain Their Limit Points

2.8 Sequential Convergence

Definition — Metric Sequential Convergence

Definition 2.8.1 (Metric Sequential Convergence)**.** Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *convergent in* X if there exists $x_0 \in X$ such that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_0) < \varepsilon \qquad \text{for all } n \geq N.$$

In this case, $\{x_n\}$ converges to x_0, written

$$x_n \to x_0, \qquad x_0 = \lim_{n\to\infty} x_n.$$

Remark (Standard quantified statement).

$$\begin{aligned} x_n \to x_0 \quad \iff \quad & x_0 \in X \ \wedge\ \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \in \mathbb{N}\colon \\ & n \geq N \implies d(x_n, x_0) < \varepsilon. \end{aligned}$$

Remark (Interpretation). A sequence converges to x_0 when its tail eventually lies inside every metric tolerance around x_0. The metric supplies the meaning of "within ε" in the ambient space X.

Remark (Historical note). This convention follows Definition 2.20 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space

Definition — Metric Neighborhood

Definition 2.8.2 (Metric Neighborhood)**.** Let (X, d) be a metric space and let $x \in X$. A subset $U \subseteq X$ is a *neighborhood of* x if there exists $\delta > 0$ such that

$$B_d(x; \delta) \subseteq U.$$

Remark (Standard quantified statement).

$$U \text{ is a neighborhood of } x \quad \iff \quad U \subseteq X \ \wedge\ \exists \delta > 0\colon B_d(x; \delta) \subseteq U.$$

Remark (Interpretation). A metric neighborhood of x is any subset of X that contains at least one open ball centered at x. The set U may be larger than that ball; the essential condition is that x has some positive-radius room inside U.

Remark (Historical note). This convention follows Definition 2.21 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball

Definition — Epsilon Neighborhood

Definition 2.8.3 (Epsilon Neighborhood)**.** Let (X, d) be a metric space, let $x \in X$, and let $\varepsilon > 0$. The *ε-neighborhood of x* is the open ball

$$B_d(x; \varepsilon) = \{y \in X : d(y, x) < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in B_d(x; \varepsilon) \quad \Longleftrightarrow \quad y \in X \ \wedge \ d(y, x) < \varepsilon.$$

Remark (Interpretation). An ε-neighborhood is the concrete neighborhood obtained by choosing a particular positive radius ε. It consists exactly of the points whose distance from x is less than that tolerance.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Metric Neighborhood

Theorem

Theorem 2.8.4 (Metric Neighborhood Criterion for Sequential Convergence)**.** *Let (X, d) be a metric space, let $\{x_n\}$ be a sequence in X, and let $x \in X$. Then*

$$x_n \to x$$

if and only if every neighborhood of x contains all but finitely many terms of $\{x_n\}$.
Go to proof.

Remark (Standard quantified statement).

$$x_n \to x \quad \Longleftrightarrow \quad \forall U \subseteq X\colon \big(U \text{ is a neighborhood of } x\big) \Longrightarrow \exists N \in \mathbb{N}\, \forall n \geq N\colon x_n \in U.$$

Remark (Interpretation). Sequential convergence can be tested by neighborhoods instead of numerical ε-bounds. A convergent sequence eventually remains inside every set that contains some open ball around the limit.

Remark (Historical note). This is the neighborhood characterization stated immediately after Definition 2.21 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Metric Neighborhood

Definition — Metric Cauchy Sequence

Definition 2.8.5 (Metric Cauchy Sequence)**.** Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *Cauchy* if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon \qquad \text{for all } n, m \geq N.$$

Remark (Standard quantified statement).

$$\{x_n\} \text{ is Cauchy} \quad \Longleftrightarrow \quad \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n, m \in \mathbb{N}\colon n, m \geq N \Longrightarrow d(x_n, x_m) < \varepsilon.$$

Remark (Interpretation). A Cauchy sequence is eventually internally stable: sufficiently late terms are close to one another, regardless of whether a limit point has already been identified in X.

Remark (Historical note). This convention follows Definition 2.22 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space

Theorem

Theorem 2.8.6 (Metric Convergent Sequences Have Unique Limits)**.** *In metric spaces, convergent sequences have unique limits.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq X\, \forall x, y \in X\colon \quad \big(x_n \to x \ \wedge \ x_n \to y\big) \Longrightarrow x = y.$$

Remark (Interpretation). In a metric space, a convergent sequence cannot settle at two different points. The positive definiteness of the metric turns zero distance between candidate limits into equality of the points.

Remark (Historical note). This result corresponds to Theorem 2.23 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Sequential Convergence

Theorem

Theorem 2.8.7 (Metric Cauchy Sequence with Convergent Subsequence). *Let $\{x_n\}$ be a Cauchy sequence in a metric space (X, d), let $x \in X$, and let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$ such that*

$$\lim_{k\to\infty} x_{n_k} = x.$$

Then $x_n \to x$.
Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq X \,\forall x \in X\colon \quad \Big(\{x_n\} \text{ is Cauchy} \\ \wedge \exists \{x_{n_k}\} \text{ subsequence of } \{x_n\}\colon x_{n_k} \to x\Big) \implies x_n \to x.$$

Remark (Interpretation). For a Cauchy sequence, one convergent subsequence determines the behavior of the whole sequence. The Cauchy condition forces the full tail to follow any subsequence tail that has already converged.

Remark (Historical note). This result corresponds to Theorem 2.24 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Cauchy Sequence
- Definition: Metric Sequential Convergence

Theorem

Theorem 2.8.8 (Coordinatewise Sequences in $\mathbb{R}^m$). *Let $\{x_n\}$ be a sequence in $\mathbb{R}^m$, and let $x_0 \in \mathbb{R}^m$. Write*

$$x_n = \big(x_n^{(1)}, \dots, x_n^{(m)}\big) \qquad \text{for all } n \in \mathbb{N} \cup \{0\}.$$

Then:

1. *$x_n \to x_0$ if and only if $x_n^{(j)} \to x_0^{(j)}$ for every $j = 1, \dots, m$.*
2. *$\{x_n\}$ is Cauchy if and only if $\{x_n^{(j)}\}$ is Cauchy for every $j = 1, \dots, m$.*

Go to proof.

Remark (Standard quantified statement).

$$x_n \to x_0 \iff \forall j \in \{1, \ldots, m\}\colon x_n^{(j)} \to x_0^{(j)},$$
$$\{x_n\} \text{ is Cauchy} \iff \forall j \in \{1, \ldots, m\}\colon \{x_n^{(j)}\} \text{ is Cauchy}.$$

Remark (Interpretation). In finite-dimensional Euclidean space, convergence and Cauchy behavior can be checked one coordinate at a time. The Euclidean metric packages the finitely many coordinate estimates into one distance estimate.

Remark (Historical note). This result corresponds to Theorem 2.25 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Metric Cauchy Sequence
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Theorem

Theorem 2.8.9 (Euclidean Cauchy Sequences Converge). *Every Cauchy sequence in $\mathbb{R}^m$ is convergent in $\mathbb{R}^m$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq \mathbb{R}^m\colon \quad \{x_n\} \text{ is Cauchy} \implies \exists x \in \mathbb{R}^m\colon x_n \to x.$$

Remark (Interpretation). $\mathbb{R}^m$ contains the limits of all of its Cauchy sequences. This is the finite-dimensional Euclidean completeness property.

Remark (Historical note). This result corresponds to Theorem 2.26 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Cauchy Sequence
- Definition: Metric Sequential Convergence
- Theorem: Coordinatewise Sequences in $\mathbb{R}^m$

Theorem

Theorem 2.8.10 (Bolzano–Weierstrass in $\mathbb{R}^m$). *Every bounded sequence in $\mathbb{R}^m$ contains a subsequence that converges in $\mathbb{R}^m$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall \{x_n\} \subseteq \mathbb{R}^m \colon \quad \{x_n\} \text{ is bounded} \Longrightarrow \exists \{x_{n_k}\} \text{ subsequence } \exists x \in \mathbb{R}^m \colon x_{n_k} \to x.$$

Remark (Interpretation). Boundedness in finite-dimensional Euclidean space is strong enough to force some subsequence to settle to a Euclidean limit.

Remark (Historical note). This result corresponds to Theorem 2.27 in Kainth's *A Comprehensive Textbook on Metric Spaces* Kainth [2023].

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Proofs

Remark (Return). Return to Lemma

Lemma (Metrics are Pseudo-Metrics)**.** *Every metric on a nonempty set X is a pseudo-metric on X. The converse is false: there exist pseudo-metrics that are not metrics.*

Professional Standard Proof.
Let d be a metric on X. The metric axioms give

$$d(x,y) \geq 0, \qquad d(x,y) = d(y,x), \qquad d(x,y) \leq d(x,z) + d(z,y)$$

for all $x, y, z \in X$. They also give $d(x,x) = 0$, since $d(x,y) = 0$ if and only if $x = y$. Hence d satisfies every pseudo-metric axiom.

For the converse, let $X = \{a, b\}$ with $a \neq b$, and define $d : X \times X \to \mathbb{R}$ by $d(x,y) = 0$ for all $x, y \in X$. Then d is nonnegative, symmetric, satisfies $d(x,x) = 0$, and satisfies the triangle inequality. Thus d is a pseudo-metric. But $d(a,b) = 0$ while $a \neq b$, so d is not a metric. □

Detailed Learning Proof.
Step 1. Compare the axioms.

A metric has nonnegativity, symmetry, and the triangle inequality. These are exactly three of the pseudo-metric requirements. The remaining pseudo-metric requirement is $d(x,x) = 0$, which follows from the metric condition

$$d(x,y) = 0 \iff x = y$$

by substituting $y = x$. Therefore every metric is a pseudo-metric.

Step 2. Build a pseudo-metric that fails to be a metric.

Take a two-point set $X = \{a, b\}$ and assign distance zero to every pair:

$$d(a,a) = d(a,b) = d(b,a) = d(b,b) = 0.$$

All pseudo-metric axioms hold. Nonnegativity and symmetry are immediate, and the triangle inequality says $0 \leq 0 + 0$, which is true.

Step 3. Identify the failed metric axiom.

The metric zero-distance condition would require

$$d(a,b) = 0 \implies a = b.$$

But $d(a,b) = 0$ and $a \neq b$. Hence this pseudo-metric is not a metric, so the converse implication is false. □

Remark (Proof structure). The forward direction is an axiom comparison. The reverse direction is disproved by a two-point zero-distance example.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Pseudo-Metric

Remark (Return). Return to Theorem

Theorem (A Norm Induces a Metric). *Let V be a real vector space and let $\|\cdot\|$ be a norm on V. Define*

$$d_{\|\cdot\|}(x, y) := \|x - y\|.$$

Then $d_{\|\cdot\|}$ is a metric on V.

Professional Standard Proof.
TODO: professional standard proof that a norm induces a metric. □

Detailed Learning Proof.
TODO: detailed learning proof that a norm induces a metric. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Norm

Remark (Return). Return to Proposition

Proposition (The Discrete Distance is a Metric). *For every nonempty set X, the discrete distance d_{disc} is a metric on X.*

Professional Standard Proof.
TODO: professional standard proof for the discrete metric. □

Detailed Learning Proof.
TODO: detailed learning proof for the discrete metric. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Discrete Metric

Remark (Return). Return to Proposition

Proposition (The Real-Line Euclidean Distance is a Metric). *The function $d_{\mathbb{R}}(x,y) = |x-y|$ is a metric on $\mathbb{R}$.*

Professional Standard Proof.
For all $x, y \in \mathbb{R}$, $d_{\mathbb{R}}(x,y) = |x-y| \geq 0$. Moreover, $|x-y| = 0$ if and only if $x - y = 0$, which holds if and only if $x = y$. Symmetry follows from

$$d_{\mathbb{R}}(x,y) = |x-y| = |-(y-x)| = |y-x| = d_{\mathbb{R}}(y,x).$$

For $x, y, z \in \mathbb{R}$, the absolute-value triangle inequality gives

$$d_{\mathbb{R}}(x,y) = |x-y| = |(x-z)+(z-y)| \leq |x-z| + |z-y| = d_{\mathbb{R}}(x,z) + d_{\mathbb{R}}(z,y).$$

Thus $d_{\mathbb{R}}$ satisfies the metric axioms. □

Detailed Learning Proof.
The metric axioms are checked directly. Nonnegativity and the zero-distance condition are properties of absolute value. Symmetry is the identity $|u| = |-u|$ applied to $u = x - y$. The triangle inequality is the ordinary absolute-value triangle inequality applied to the decomposition

$$x - y = (x-z) + (z-y).$$

□

Remark (Proof structure). The proof translates each metric axiom into a standard property of absolute value.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}$

Remark (Return). Return to Theorem

Theorem (Cauchy–Schwarz Inequality). *For every $x, y \in \mathbb{R}^n$,*

$$|x \cdot y| \leq |x|\,|y|.$$

Equality holds if and only if x and y are linearly dependent.

Professional Standard Proof.
If $y = 0$, then both sides are 0, and the equality condition is immediate. Assume $y \neq 0$. For every $t \in \mathbb{R}$,

$$0 \leq |x - ty|^2 = |x|^2 - 2t(x \cdot y) + t^2|y|^2.$$

Choose $t = (x \cdot y)/|y|^2$. Substitution gives

$$0 \leq |x|^2 - \frac{(x \cdot y)^2}{|y|^2}.$$

Hence

$$(x \cdot y)^2 \leq |x|^2|y|^2,$$

and taking square roots gives $|x \cdot y| \leq |x|\,|y|$.

Equality holds exactly when the chosen square has value 0, namely when

$$x - \frac{x \cdot y}{|y|^2}y = 0.$$

Thus equality implies that x is a scalar multiple of y, so x and y are linearly dependent. Conversely, if $x = \lambda y$, then

$$|x \cdot y| = |\lambda|\,|y|^2 = |\lambda y|\,|y| = |x|\,|y|,$$

so equality holds. This also covers the case in which one vector is zero. □

Detailed Learning Proof.
The key observation is that squared length is always nonnegative. For any real number t, the vector $x - ty$ therefore satisfies

$$0 \leq |x - ty|^2.$$

Expanding this square with the dot product gives a quadratic expression in t:

$$|x - ty|^2 = |x|^2 - 2t(x \cdot y) + t^2|y|^2.$$

When $y \neq 0$, the choice $t = (x \cdot y)/|y|^2$ minimizes this quadratic. Since the minimum is still nonnegative,

$$|x|^2 - \frac{(x \cdot y)^2}{|y|^2} \geq 0.$$

Multiplying by $|y|^2$ gives

$$(x \cdot y)^2 \leq |x|^2|y|^2.$$

The desired inequality follows by taking square roots.

The equality case occurs precisely when the minimized square is zero. That means $x - ty = 0$ for the minimizing value of t, so $x = ty$. Hence the two vectors are linearly dependent. If two vectors are linearly dependent, one is a scalar multiple of the other, and direct substitution gives equality. $\square$

Remark (Proof structure). The proof minimizes the nonnegative quadratic $t \mapsto |x - ty|^2$. The inequality comes from the minimum being nonnegative; the equality condition comes from the minimum being zero.

Remark (Dependencies).

- Definition: Euclidean Space
- Definition: Euclidean Distance on $\mathbb{R}^n$

Remark (Return). Return to Corollary

Corollary (Minkowski's Inequality). *For every $x, y \in \mathbb{R}^n$,*

$$|x + y| \leq |x| + |y|.$$

Professional Standard Proof.
Using the dot product and Cauchy–Schwarz,

$$\begin{aligned} |x+y|^2 &= (x+y) \cdot (x+y) \\ &= |x|^2 + 2x \cdot y + |y|^2 \\ &\leq |x|^2 + 2|x|\,|y| + |y|^2 \\ &= (|x| + |y|)^2. \end{aligned}$$

Both sides are nonnegative, so taking square roots gives

$$|x + y| \leq |x| + |y|.$$

Writing the norms in coordinates gives the displayed coordinate form. □

Detailed Learning Proof.
The Euclidean norm satisfies

$$|x + y|^2 = (x + y) \cdot (x + y).$$

Expanding the dot product gives

$$|x + y|^2 = |x|^2 + 2x \cdot y + |y|^2.$$

Cauchy–Schwarz gives $x \cdot y \leq |x|\,|y|$, so

$$|x + y|^2 \leq |x|^2 + 2|x|\,|y| + |y|^2 = (|x| + |y|)^2.$$

Since lengths are nonnegative, the square-root operation preserves this inequality. Hence $|x + y| \leq |x| + |y|$. □

Remark (Proof structure). The proof expands the squared length of $x + y$, bounds the cross term by Cauchy–Schwarz, and then takes square roots.

Remark (Dependencies).

- Theorem: Cauchy–Schwarz Inequality
- Definition: Euclidean Distance on $\mathbb{R}^n$

Remark (Return). Return to Proposition

Proposition (The Euclidean Distance on $\mathbb{R}^n$ is a Metric). *The function d_2 is a metric on $\mathbb{R}^n$.*

Professional Standard Proof.
For $x, y \in \mathbb{R}^n$,

$$d_2(x,y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \geq 0.$$

This distance is 0 if and only if every square $(x_i - y_i)^2$ is 0, which is equivalent to $x_i = y_i$ for every i, hence to $x = y$. Symmetry follows from $(x_i - y_i)^2 = (y_i - x_i)^2$.

For the triangle inequality, set $u = x - z$ and $v = z - y$. Then

$$d_2(x,y) = |u+v|, \qquad d_2(x,z) = |u|, \qquad d_2(z,y) = |v|.$$

Using Cauchy–Schwarz,

$$|u+v|^2 = |u|^2 + 2u \cdot v + |v|^2 \leq |u|^2 + 2|u|\,|v| + |v|^2 = (|u| + |v|)^2.$$

Both sides are nonnegative, so $|u+v| \leq |u| + |v|$. Therefore

$$d_2(x,y) \leq d_2(x,z) + d_2(z,y).$$

Thus d_2 is a metric. □

Detailed Learning Proof.
The first three metric requirements are coordinate checks. A sum of squares is nonnegative, and it vanishes exactly when every coordinate difference vanishes. Symmetry holds because changing $x_i - y_i$ to $y_i - x_i$ only changes the sign before squaring.

The triangle inequality is the vector-length statement

$$|u+v| \leq |u| + |v|.$$

After squaring, it becomes

$$|u+v|^2 \leq (|u| + |v|)^2.$$

Expanding the left side gives $|u|^2 + 2u \cdot v + |v|^2$. Cauchy–Schwarz gives $u \cdot v \leq |u|\,|v|$, so the desired squared inequality follows. Taking square roots preserves the inequality because both sides are nonnegative. □

Remark (Proof structure). The proof verifies nonnegativity, zero-distance, and symmetry by coordinates. The triangle inequality is Minkowski's inequality in finite dimensions, proved from Cauchy–Schwarz.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Theorem: Cauchy–Schwarz Inequality

Remark (Return). Return to Corollary

Corollary (Euclidean Metric). *For $x, y \in \mathbb{R}^n$, define*

$$d_2(x, y) = \sqrt{\sum_{i=1}^{n} |x_i - y_i|^2}.$$

Then $(\mathbb{R}^n, d_2)$ is a metric space.

Professional Standard Proof.
The function d_2 is the Euclidean distance on $\mathbb{R}^n$. For all $x, y \in \mathbb{R}^n$, $d_2(x, y) \geq 0$, and $d_2(x, y) = 0$ if and only if each coordinate difference $x_i - y_i$ is 0, equivalently $x = y$. Symmetry follows from $|x_i - y_i| = |y_i - x_i|$ for every coordinate.

For the triangle inequality, write

$$x - y = (x - z) + (z - y).$$

Applying Minkowski's inequality gives

$$d_2(x, y) \leq d_2(x, z) + d_2(z, y).$$

Thus d_2 satisfies the metric axioms, so $(\mathbb{R}^n, d_2)$ is a metric space. □

Detailed Learning Proof.
The formula for d_2 is the length of the vector $x - y$. A length is nonnegative. It is zero exactly when every coordinate of $x - y$ is zero, which means $x = y$. It is symmetric because replacing $x - y$ by $y - x$ only changes signs before absolute values and squares are taken.

The only structural point is the triangle inequality. The displacement from y to x can be split through z:

$$x - y = (x - z) + (z - y).$$

Minkowski's inequality says that the length of a sum is at most the sum of the lengths. Therefore the direct distance $d_2(x, y)$ is at most $d_2(x, z) + d_2(z, y)$. □

Remark (Proof structure). The proof checks the easy metric axioms coordinatewise and uses Minkowski's inequality for the triangle inequality.

Remark (Dependencies).

- Definition: Metric Space
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Minkowski's Inequality

Remark (Return). Return to Proposition

Proposition (Taxicab Metric). *The function d_1 is a metric on $\mathbb{R}^n$.*

Professional Standard Proof.
TODO: professional standard proof for the taxicab metric. □

Detailed Learning Proof.
TODO: detailed learning proof for the taxicab metric. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Taxicab Metric

Remark (Return). Return to Theorem

Theorem (d_p is a Metric)**.** *For every* $1 \leq p < \infty$*, the function* d_p *is a metric on* $\mathbb{R}^n$*.*

Professional Standard Proof.
TODO: professional standard proof for the p-metric. □

Detailed Learning Proof.
TODO: detailed learning proof for the p-metric. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: p-Metric on $\mathbb{R}^n$

Remark (Return). Return to Proposition

Proposition (The Complex Plane is a Metric Space)**.** *The pair* $(\mathbb{C}, d_{\mathbb{C}})$ *is a metric space.*

Professional Standard Proof.
TODO: professional standard proof for the complex plane as a metric space. □

Detailed Learning Proof.
TODO: detailed learning proof for the complex plane as a metric space. □

Remark (Proof structure)*.*

Remark (Dependencies)*.*

- Definition: Metric Space
- Definition: Complex Metric
- Corollary: Euclidean Metric

Remark (Return). Return to Proposition

Proposition (The Supremum Distance on Bounded Sequences is a Metric). *The function d_∞ is a metric on $\ell^\infty(\mathbb{R})$.*

Professional Standard Proof.
TODO: professional standard proof for the supremum metric on bounded real sequences. □

Detailed Learning Proof.
TODO: detailed learning proof for the supremum metric on bounded real sequences. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Bounded Real Sequence Space

Remark (Return). Return to Theorem

Theorem (Metric Open Set Closure Properties)**.** *Let X be a metric space. Then the empty set $\varnothing$ and the space X are open sets, an arbitrary union of open subsets of X is open, and any finite intersection of open subsets of X is open.*

Professional Standard Proof.
TODO: professional standard proof for metric-open-set-closure-properties. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-open-set-closure-properties. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Open Set
- Definition: Metric Interior Point

Remark (Return). Return to Theorem

Theorem (Open Balls are Open). *Open balls in metric spaces are open sets.*

Professional Standard Proof.
TODO: professional standard proof for metric-open-balls-are-open. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-open-balls-are-open. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Open Ball
- Definition: Metric Open Set

Remark (Return). Return to Theorem

Theorem (Interior is the Largest Open Subset)**.** *Let E be a subset of a metric space X. Then E° is the largest open subset of X contained in E.*

Professional Standard Proof.
TODO: professional standard proof for metric-interior-largest-open-subset. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-interior-largest-open-subset. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Interior
- Definition: Metric Open Set
- Theorem: Open Balls are Open

Remark (Return). Return to Theorem

Theorem (Closed Sets Contain Their Limit Points). *Let X be a metric space and let $E \subseteq X$. Then E is closed if and only if $E' \subseteq E$.*

Professional Standard Proof.
TODO: professional standard proof for metric-closed-sets-contain-limit-points. ☐

Detailed Learning Proof.
TODO: detailed learning proof for metric-closed-sets-contain-limit-points. ☐

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Closed Set
- Definition: Metric Limit Point and Isolated Point

Remark (Return). Return to Theorem

Theorem (Sequential Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if there exists a sequence of distinct terms in E that converges to x.*

Professional Standard Proof.
TODO: professional standard proof for metric-limit-point-sequential-characterization. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-limit-point-sequential-characterization. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Limit Point and Isolated Point
- Definition: Metric Sequential Convergence

Remark (Return). Return to Corollary

Corollary (Neighborhood Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if every neighborhood of x contains infinitely many points of E.*

Professional Standard Proof.
TODO: professional standard proof for metric-limit-point-neighborhood-characterization. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-limit-point-neighborhood-characterization. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Neighborhood
- Theorem: Sequential Characterization of Limit Points

Remark (Return). Return to Theorem

Theorem (Bolzano–Weierstrass for Bounded Infinite Subsets)**.** *Every bounded infinite subset of $\mathbb{R}^m$ has a limit point in $\mathbb{R}^m$.*

Professional Standard Proof.
TODO: professional standard proof for metric-bolzano-weierstrass-bounded-infinite-subsets. ∎

Detailed Learning Proof.
TODO: detailed learning proof for metric-bolzano-weierstrass-bounded-infinite-subsets. ∎

Remark (Proof structure).

Remark (Dependencies).

- Theorem: Bolzano–Weierstrass in $\mathbb{R}^m$
- Theorem: Sequential Characterization of Limit Points

Remark (Return). Return to Proposition

Proposition (The Derived Set is Closed). *The set of limit points of any set is closed.*

Professional Standard Proof.
TODO: professional standard proof for metric-derived-set-closed. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-derived-set-closed. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Limit Point and Isolated Point
- Theorem: Closed Sets Contain Their Limit Points

Remark (Return). Return to Theorem

Theorem (Metric Neighborhood Criterion for Sequential Convergence)**.** *Let (X, d) be a metric space, let $\{x_n\}$ be a sequence in X, and let $x \in X$. Then $x_n \to x$ if and only if every neighborhood of x contains all but finitely many terms of $\{x_n\}$.*

Professional Standard Proof.
TODO: professional standard proof for metric-neighborhood-criterion-sequential-convergence. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-neighborhood-criterion-sequential-convergence. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Metric Neighborhood
- Definition: Open Ball

Remark (Return). Return to Theorem

Theorem (Metric Convergent Sequences Have Unique Limits)**.** *In metric spaces, convergent sequences have unique limits.*

Professional Standard Proof.
TODO: professional standard proof for metric-convergent-sequences-unique-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-convergent-sequences-unique-limits. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Sequential Convergence

Remark (Return). Return to Theorem

Theorem (Metric Cauchy Sequence with Convergent Subsequence). *Let $\{x_n\}$ be a Cauchy sequence in a metric space (X, d), let $x \in X$, and let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$ such that $\lim_{k\to\infty} x_{n_k} = x$. Then $x_n \to x$.*

Professional Standard Proof.
TODO: professional standard proof for metric-cauchy-sequence-convergent-subsequence. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-cauchy-sequence-convergent-subsequence. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Space
- Definition: Metric Cauchy Sequence
- Definition: Metric Sequential Convergence

Remark (Return). Return to Theorem

Theorem (Coordinatewise Sequences in $\mathbb{R}^m$). *Let $\{x_n\}$ be a sequence in $\mathbb{R}^m$, and let $x_0 \in \mathbb{R}^m$. Write*

$$x_n = \left(x_n^{(1)}, \ldots, x_n^{(m)}\right) \qquad \textit{for all } n \in \mathbb{N} \cup \{0\}.$$

Then $x_n \to x_0$ if and only if $x_n^{(j)} \to x_0^{(j)}$ for every $j = 1, \ldots, m$, and $\{x_n\}$ is Cauchy if and only if $\{x_n^{(j)}\}$ is Cauchy for every $j = 1, \ldots, m$.

Professional Standard Proof.
TODO: professional standard proof for metric-euclidean-coordinatewise-sequences. ☐

Detailed Learning Proof.
TODO: detailed learning proof for metric-euclidean-coordinatewise-sequences. ☐

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Metric Cauchy Sequence
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Remark (Return). Return to Theorem

Theorem (Euclidean Cauchy Sequences Converge). *Every Cauchy sequence in $\mathbb{R}^m$ is convergent in $\mathbb{R}^m$.*

Professional Standard Proof.
TODO: professional standard proof for metric-euclidean-cauchy-sequences-converge. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-euclidean-cauchy-sequences-converge. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Cauchy Sequence
- Definition: Metric Sequential Convergence
- Theorem: Coordinatewise Sequences in $\mathbb{R}^m$

Remark (Return). Return to Theorem

Theorem (Bolzano–Weierstrass in $\mathbb{R}^m$)**.** *Every bounded sequence in $\mathbb{R}^m$ contains a subsequence that converges in $\mathbb{R}^m$.*

Professional Standard Proof.
TODO: professional standard proof for metric-euclidean-bolzano-weierstrass. □

Detailed Learning Proof.
TODO: detailed learning proof for metric-euclidean-bolzano-weierstrass. □

Remark (Proof structure).

Remark (Dependencies).

- Definition: Metric Sequential Convergence
- Definition: Euclidean Distance on $\mathbb{R}^n$
- Corollary: Euclidean Metric

Chapter 3

Topology

3.1 Topology

3.2 Definitions

Proofs

Chapter 4

Set Algebra

SERIES	VOLUME IV	BOOK I	CHAPTER 4
From Cantor to Ito	Mathematical Spaces	Mathematical Spaces	Set Algebra

4.1 Set Algebras

4.2 Systems of Sets

4.2.1 Systems of Sets

Comparison of Set Systems

Property	Semiring	Ring	Algebra	σ-algebra	Topology
Contains $\varnothing$	✓	✓	✓	✓	✓
Contains X	—	—	✓	✓	✓
Closed under $\cap$ (finite)	✓	✓	✓	✓	✓
Closed under $\cup$ (finite)	—	✓	✓	✓	✓
Closed under $\cup$ (countable)	—	—	—	✓	—
Closed under $\cup$ (arbitrary)	—	—	—	—	✓
Closed under complement	—	—	✓	✓	—
Difference $A \setminus B$	finite disj. union	✓	✓	✓	—

The definitions and theorems of the previous sections have been about individual sets and their operations. A second level of structure emerges when one asks: which *collections* of sets are closed under given operations? This question is not abstract generality for its own sake. The three subjects that depend most heavily on this chapter — topology, measure theory, and functional analysis — each begin by specifying a particular type of collection on a given set X. A topology specifies which subsets are "open." A σ-algebra specifies which subsets are "measurable." In both cases, the definition is precisely a list of closure conditions: which set operations applied to members of the collection must produce another member.

Mastering these four structures here, as purely set-theoretic objects, means that when topology and measure theory are encountered, all the foundational work is already done. The task will be to understand the *geometric* or *analytic* meaning of the axioms, not to struggle with the set-theoretic mechanics.

Definition (Semiring of Sets)

Definition 4.2.1 (Semiring of Sets)**.** A collection $\mathcal{S}$ of subsets of a set X is a *semiring* if:

(i) $\varnothing \in \mathcal{S}$;

(ii) $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$;

(iii) for any $A, B \in \mathcal{S}$, the difference $A \setminus B$ is a *finite disjoint union* of members of $\mathcal{S}$: there exist pairwise disjoint $C_1, \ldots, C_n \in \mathcal{S}$ with $A \setminus B = C_1 \sqcup \cdots \sqcup C_n$.

Remark (Dependencies).

- Subset
- Empty Set
- Intersection
- Set Difference
- Union
- Pairwise Disjoint Family
- Finite and Infinite Sets

Example (Half-open intervals form a semiring). Let $X = \mathbb{R}$ and let $\mathcal{S}$ be the collection of all half-open intervals $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$, together with $\varnothing$. Then $\mathcal{S}$ is a semiring:

- $[a, b) \cap [c, d) = [\max(a, c), \min(b, d))$, which is again a half-open interval (or $\varnothing$);
- $[a, b) \setminus [c, d)$ is either $\varnothing$, $[a, c)$, $[d, b)$, or $[a, c) \sqcup [d, b)$ — in every case, a finite disjoint union of half-open intervals.

This semiring is the natural starting point for constructing Lebesgue measure: one first defines length on $\mathcal{S}$ (as $b - a$ for $[a, b)$), then extends to the generated ring and σ-algebra.

Remark (Why semirings appear in measure theory). Lebesgue measure is most naturally defined first on a semiring, because semirings have enough structure to state the countable additivity condition for a *premeasure*, yet are simple enough that explicit formulas for sets like $[a, b)$ are easy to write down. The general extension theorem (Carathéodory's theorem) then extends the premeasure to the full σ-algebra. This is why Kolmogorov & Fomin introduce semirings before algebras: they are the base case of the extension hierarchy.

Definition (Ring of Sets)

Definition 4.2.2 (Ring of Sets)**.** A collection $\mathcal{R}$ of subsets of X is a *ring of sets* if:

(i) $\varnothing \in \mathcal{R}$;

(ii) $A, B \in \mathcal{R}$ implies $A \cup B \in \mathcal{R}$;

(iii) $A, B \in \mathcal{R}$ implies $A \setminus B \in \mathcal{R}$.

Remark (Dependencies).

- Subset
- Empty Set
- Union
- Set Difference

Remark (Derived closure properties of a ring). A ring is closed under finitely many of each operation. Specifically, if $\mathcal{R}$ is a ring then:

- $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$ (intersection follows from set difference);
- $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{R}$ (symmetric difference follows from union and difference).

A ring is closed under all finite Boolean combinations of its elements.

Remark (Why "ring"). The name comes from algebra. Under the operations of symmetric difference $\triangle$ (playing the role of addition) and intersection $\cap$ (multiplication), a ring of sets is a commutative ring in the algebraic sense: $\triangle$ is commutative and associative with identity $\varnothing$, every element is its own inverse ($A \triangle A = \varnothing$), and $\cap$ distributes over $\triangle$.

Definition (Algebra of Sets)

Definition 4.2.3 (Algebra of Sets)**.** A collection $\mathcal{A}$ of subsets of X is an *algebra of sets* (or *Boolean algebra of sets*, or *field of sets*) if:

(i) $X \in \mathcal{A}$;

(ii) $A \in \mathcal{A}$ implies $A^c = X \setminus A \in \mathcal{A}$;

(iii) $A, B \in \mathcal{A}$ implies $A \cup B \in \mathcal{A}$.

Remark (Dependencies).

- Subset
- Complement
- Union

Remark (Derived properties of an algebra). An algebra is a ring that also contains X. Since $\varnothing = X^c$ and $X \in \mathcal{A}$, every algebra contains $\varnothing$. Closure under finite intersection follows: $A \cap B = (A^c \cup B^c)^c \in \mathcal{A}$ by De Morgan. An algebra is therefore closed under all finite Boolean operations: finite unions, finite intersections, complements, and differences.

Definition (σ-Algebra)

Definition 4.2.4 (σ-Algebra)**.** A collection $\mathcal{M}$ of subsets of X is a *σ-algebra* (or *σ-field*) on X if:

(i) $X \in \mathcal{M}$;

(ii) $A \in \mathcal{M}$ implies $A^c \in \mathcal{M}$;

(iii) if $\{A_n\}_{n\in\mathbb{N}}$ is a countable family in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is called a *measurable space.*

Remark (Dependencies).

- Subset
- Empty Set
- Complement
- Union
- Countable Set

Remark (Derived closure properties of a σ-algebra). Every σ-algebra is an algebra (take the union of a finite family by padding with $\varnothing$). But a σ-algebra is additionally closed under countable intersections: by De Morgan,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c\right)^c \in \mathcal{M}.$$

The σ in "σ-algebra" is notation for "countable" (from the German *Summe*), signalling the step up from finite to countable closure.

Remark (Why countable and not arbitrary). A σ-algebra is closed under countable unions but not arbitrary ones. This is not a deficiency — it is the right condition for measure theory. Measure is defined to be countably additive: if $\{A_n\}$ is a countable disjoint family, then $\mu\big(\bigcup A_n\big) = \sum \mu(A_n)$. Extending to uncountable unions would force every set to have either measure zero or infinite measure (a useless theory). The countability restriction is what makes the theory nontrivial.

By contrast, a topology (defined below) is closed under *arbitrary* unions but only *finite* intersections. This is a different trade-off: open sets are defined by local conditions, and any union of open sets is open because being open is a pointwise property. Countable intersections of open sets are *not* required to be open; those are a separate class called G_δ sets.

Definition (Topology)

Definition 4.2.5 (Topology)**.** A *topology* on a set X is a collection τ of subsets of X satisfying:

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) if $\{U_\alpha\}_{\alpha \in I}$ is any family of sets in τ (indexed by any set I), then $\bigcup_{\alpha \in I} U_\alpha \in \tau$;

(iii) if $U_1, \ldots, U_n \in \tau$ (finitely many), then $U_1 \cap \cdots \cap U_n \in \tau$.

The pair (X, τ) is a *topological space*. Members of τ are called *open sets*.

Remark (Dependencies).

- Subset
- Empty Set
- Indexed Family of Sets
- Indexed Union
- Indexed Intersection
- Finite and Infinite Sets

Remark (Why arbitrary unions but only finite intersections)*.* A topology formalizes the idea of "nearness" without a distance function. The condition that arbitrary unions of open sets are open reflects the following: if a point x is near some set, and that set is a union of smaller sets, then x is near one of those smaller sets. This is a purely local condition and works for any union.

Finite intersections must be open because: if x lies in the interior of U_1 and also in the interior of U_2, it lies in the interior of $U_1 \cap U_2$. This argument works for finitely many sets (take the smallest neighborhood). For a countably infinite intersection, the "smallest neighborhood" might shrink to nothing — consider $\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\}$ in $\mathbb{R}$, which is not open. So infinite intersections of open sets are not required to be open.

Theorem (Intersection of σ-algebras is a σ-algebra)

Theorem 4.2.6 (Intersection of σ-algebras is a σ-algebra)**.** *Let $\{\mathcal{M}_\alpha\}_{\alpha\in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} \;:=\; \bigcap_{\alpha\in I} \mathcal{M}_\alpha$$

is a σ-algebra on X. *Go to proof.*

Remark (Standard quantified statement). $\big(\forall\alpha \in I,\ \mathcal{M}_\alpha$ is a σ-algebra on $X\big) \;\to\; \bigcap_{\alpha\in I}\mathcal{M}_\alpha$ is a σ-algebra Explicitly: $X \in \bigcap\mathcal{M}_\alpha$; $A \in \bigcap\mathcal{M}_\alpha \to A^c \in \bigcap\mathcal{M}_\alpha$; $\forall n,\ A_n \in \bigcap\mathcal{M}_\alpha \to \bigcup_n A_n \in \bigcap\mathcal{M}_\alpha$.

Remark (Dependencies).

- σ-Algebra
- Indexed Family of Sets
- Intersection

Remark (Proof). Check each axiom: (i) $X \in \mathcal{M}_\alpha$ for all α, so $X \in \bigcap_\alpha \mathcal{M}_\alpha$. (ii) If $A \in \mathcal{M}$ then $A \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}_\alpha$ for all α, so $A^c \in \mathcal{M}$. (iii) If $\{A_n\} \subseteq \mathcal{M}$ then $\{A_n\} \subseteq \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}_\alpha$ for all α, so $\bigcup A_n \in \mathcal{M}$. All three conditions are verified. The same argument works for algebras, rings, and topologies (with appropriate closure conditions).

Definition (Generated σ-Algebra)

Definition 4.2.7 (Generated σ-Algebra)**.** Let $\mathcal{F}$ be any collection of subsets of X. The *σ-algebra generated by $\mathcal{F}$*, written $\sigma(\mathcal{F})$, is the smallest σ-algebra on X that contains $\mathcal{F}$:

$$\sigma(\mathcal{F}) \;:=\; \bigcap\big\{\,\mathcal{M} \;\big|\; \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F}\subseteq\mathcal{M}\,\big\}.$$

The collection $\mathcal{F}$ is called a *generating family* for $\sigma(\mathcal{F})$.

Remark (Dependencies).

- σ-Algebra
- Subset
- Intersection
- Intersection of σ-Algebras

Remark (Why the definition is valid). The set of σ-algebras on X containing $\mathcal{F}$ is nonempty: $\mathcal{P}(X)$ is a σ-algebra containing every subset of X. Theorem 4.2.6 guarantees their intersection is again a σ-algebra. So $\sigma(\mathcal{F})$ is always well-defined. The same construction works for generated algebras, generated rings, and (with a small modification for topologies) generated topologies.

Remark (How to think about generated structures). $\sigma(\mathcal{F})$ contains $\mathcal{F}$, and then contains everything that must be present in any σ-algebra that contains $\mathcal{F}$: complements of elements of $\mathcal{F}$, countable unions of elements of $\mathcal{F}$, countable unions of complements of elements of $\mathcal{F}$, and so on — iterating indefinitely. The generated σ-algebra is the "closure" of $\mathcal{F}$ under all σ-algebraic operations. In practice, one specifies $\mathcal{F}$ as a convenient collection (e.g., intervals) and then works with the full σ-algebra by knowing which operations are allowed.

Definition (Borel σ-Algebra)

Definition 4.2.8 (Borel σ-Algebra)**.** Let (X, τ) be a topological space. The *Borel σ-algebra* on X, written $\mathcal{B}(X)$, is the σ-algebra generated by the open sets:

$$\mathcal{B}(X) \ := \ \sigma(\tau).$$

Members of $\mathcal{B}(X)$ are called *Borel sets*.

Remark (Dependencies).

- Topology
- Generated σ-Algebra

Example (F_σ and G_δ sets). In a topological space X:

- A G_δ *set* is a countable intersection of open sets: $A = \bigcap_{n=1}^{\infty} U_n$ with each U_n open.
- An F_σ *set* is a countable union of closed sets: $A = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed.

These names come from German: G for *Gebiet* (open set), δ for *Durchschnitt* (intersection); F for *Ferme* (closed), σ for *Summe* (union). Both G_δ and F_σ sets are Borel sets. They are dual: A is G_δ if and only if A^c is F_σ.

These classes appear in analysis and measure theory as the natural "next level" above open and closed sets. In $\mathbb{R}$: the set of continuity points of a monotone function is a G_δ set; the rational numbers $\mathbb{Q}$ are an F_σ set; the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are a G_δ set.

Remark (Transition). The four structures — semiring, ring, algebra, σ-algebra — and the topology form a hierarchy of increasing closure strength. Measure theory builds from the bottom (semiring, ring) upward; topology is its own branch with a different closure pattern. The Borel σ-algebra bridges the two: it is generated by the topology, so it captures all sets describable by topological operations, and it is the standard "default" σ-algebra in any analytic context. Whenever a later chapter says "let f be a measurable function" without specifying the σ-algebra, it almost certainly means measurability with respect to the Borel σ-algebra.

4.3 The Power Set and Characteristic Functions

4.3.1 The Power Set and Characteristic Functions

Proofs

Remark (Return). Return to Theorem

Theorem (Intersection of -algebras is a -algebra). *Let $\{\mathcal{M}_\alpha\}_{\alpha \in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} := \bigcap_{\alpha \in I} \mathcal{M}_\alpha$$

is a σ-algebra on X.

Professional Standard Proof.
TODO: professional standard proof for sigma-intersection. □

Detailed Learning Proof.
TODO: detailed learning proof for sigma-intersection. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Chapter 5

Algebras of Sets

SERIES	VOLUME IV	BOOK I	CHAPTER 5
From Cantor to Ito	Mathematical Spaces	Mathematical Spaces	Algebras of Sets

5.1 Set Operations and Hierarchy

5.2 Boolean Algebra of Sets

5.3 Rings of Sets

Toolkit — Boolean Ring Examples

Concept family: Concrete Boolean ring examples and structural constraints.

Atomic items in this section:

- Remark: No Boolean ring of order 3 exists
- Example: The four-element Boolean ring $\mathbf{2}^2$
- Remark: Boolean rings have order 2^n

Key constraint: $(R, +)$ in any Boolean ring is an elementary abelian 2-group, so $|R| = 2^n$ for some $n \geq 0$.

Toolkit — Boolean Rings

Concept family: Boolean rings (Halmos–Givant §1) and their immediate consequences from the single idempotence axiom.

Atomic items in this section:

- Definition: Boolean Ring
- Definition: Idempotent Element
- Definition: Characteristic 2
- Lemma: Self Additive Inverse ($p + p = 0$)
- Lemma: Self Negation ($-p = p$)

Key predicate:

$$\mathrm{BooleanRing}(R, +, \cdot, -, 0, 1) \;:\equiv\; \mathrm{Ring}(R, +, \cdot, -, 0, 1) \;\wedge\; \mathrm{IdempotentRing}(R, \cdot).$$

Central theorem chain: Idempotence $\Rightarrow$ Characteristic 2 $\Rightarrow$ Self Negation ($-p = p$) $\Rightarrow$ Commutativity of $\cdot$.

Toolkit — Rings of Sets

Concept family: The ring axioms (Halmos–Givant §1) as the algebraic foundation for Boolean algebras of sets.

Atomic items in this section:

- Definition: Ring

Key predicate:

$$\mathrm{Ring}(R, +, \cdot, -, 0, 1) \;:\equiv\; \mathrm{AbelianGroup}(R, +, -, 0) \wedge \mathrm{Monoid}(R, \cdot, 1) \wedge \mathrm{Distributes}(\cdot, +, R). \blacksquare$$

Axiom numbering follows Halmos–Givant; axiom (4) (commutativity of multiplication) is not a ring axiom but a theorem in the Boolean setting.

Definition — Ring

Definition 5.3.1 (Ring)**.** A **ring** is a non-empty set R together with two binary operations $+$ and $\cdot$ on R, a unary operation $-$ on R, and two distinguished elements $0, 1 \in R$, satisfying the following axioms for all $p, q, r \in R$:

(1) $p + (q + r) = (p + q) + r$ *(additive associativity)*

(2) $p \cdot (q \cdot r) = (p \cdot q) \cdot r$ *(multiplicative associativity)*

(3) $p + q = q + p$ *(additive commutativity)*

(5) $p + 0 = p$ *(additive identity)*

(6) $p \cdot 1 = p$ *(multiplicative identity)*

(7) $p + (-p) = 0$ *(additive inverse)*

(8) $p \cdot (q + r) = p \cdot q + p \cdot r$ *(left distributivity)*

(9) $(q + r) \cdot p = q \cdot p + r \cdot p$ *(right distributivity)*

Axiom numbering follows Halmos–Givant Givant and Halmos [2009]; axiom (4) is commutativity of multiplication, which is not a ring axiom but a theorem in the Boolean setting. A ring possessing a distinguished element 1 satisfying axiom (6) is called a **ring with unit**. A ring satisfying $p \cdot q = q \cdot p$ for all $p, q \in R$ is called a **commutative ring**.

Remark (Standard quantified statement).

$$\begin{aligned}
(R, +, \cdot, -, 0, 1) \text{ is a ring } \iff \forall p, q, r \in R:&\\
&p + (q + r) = (p + q) + r\\
\wedge \quad &p \cdot (q \cdot r) = (p \cdot q) \cdot r\\
\wedge \quad &p + q = q + p\\
\wedge \quad &p + 0 = p\\
\wedge \quad &p \cdot 1 = p\\
\wedge \quad &p + (-p) = 0\\
\wedge \quad &p \cdot (q + r) = p \cdot q + p \cdot r\\
\wedge \quad &(q + r) \cdot p = q \cdot p + r \cdot p.
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Ring}(R, +, \cdot, -, 0, 1) \iff \mathrm{AbelianGroup}(R, +, -, 0) \wedge \mathrm{Monoid}(R, \cdot, 1) \wedge \mathrm{Distributes}(\cdot, +, R) \blacksquare$$

Remark (Negated quantified statement).

$$
\begin{aligned}
(R,+,\cdot,-,0,1) \text{ is } \textbf{not} \text{ a ring } \iff \exists p,q,r \in R:& \\
& p+(q+r) \neq (p+q)+r \\
\vee \quad & p\cdot(q\cdot r) \neq (p\cdot q)\cdot r \\
\vee \quad & p+q \neq q+p \\
\vee \quad & p+0 \neq p \\
\vee \quad & p\cdot 1 \neq p \\
\vee \quad & p+(-p) \neq 0 \\
\vee \quad & p\cdot(q+r) \neq p\cdot q + p\cdot r \\
\vee \quad & (q+r)\cdot p \neq q\cdot p + r\cdot p.
\end{aligned}
$$

Remark (Failure modes). The failure modes are exactly the ways that one of the defining structural conditions can fail.

Remark (Failure mode decomposition). The eight axioms partition into three structural groups:

$$
\underbrace{(1),(3),(5),(7)}_{\text{AbelianGroup}(R,+,-,0)} \qquad \underbrace{(2),(6)}_{\text{Monoid}(R,\cdot,1)} \qquad \underbrace{(8),(9)}_{\text{Distributes}(\cdot,+,R)}
$$

A structure violating any axiom in the first group lacks a coherent additive group. A structure violating any axiom in the second group lacks a multiplicative identity or associativity. A structure violating the third group possesses two individually well-behaved operations that do not interact coherently; distributivity is the sole structural bridge between $+$ and $\cdot$, and its failure is sufficient to destroy the ring property.

Remark (Interpretation). A ring abstracts the arithmetic of the integers. The additive structure is always a commutative group. The multiplicative structure is a monoid — associative with unit — but need not be commutative; axiom (4) of Halmos–Givant, commutativity of multiplication, is absent from the ring axioms. In the Boolean setting, multiplicative commutativity is not assumed but derived as a theorem from idempotence alone. Distributivity encodes the coherent interaction between the two operations: it is the axiom that makes $\cdot$ and $+$ form a single algebraic system rather than two independent ones.

Definition — Boolean Ring

Definition 5.3.2 (Boolean Ring)**.** A **Boolean ring** is an idempotent ring with unit. That is, a Boolean ring is a ring $(R, +, \cdot, -, 0, 1)$ satisfying the additional axiom:

$$p \cdot p = p \qquad \text{for all } p \in R. \tag{11}$$

An element p satisfying $p \cdot p = p$ is called **idempotent**. A ring in which every element is idempotent is called an **idempotent ring**.

The official axiom set for a Boolean ring consists of axioms (1)–(3), (5), (6), (8)–(11), where axiom (7) (the additive inverse law) is replaced by axiom (10) (characteristic 2), which is derived as a theorem. Axiom numbering follows Halmos–Givant Givant and Halmos [2009].

Remark (Standard quantified statement)*.*

$$\text{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \text{Ring}(R, +, \cdot, -, 0, 1) \;\wedge\; \forall p \in R\colon p \cdot p = p.$$

Remark (Definition predicate reading)*.*

$$\text{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \text{Ring}(R, +, \cdot, -, 0, 1) \;\wedge\; \text{IdempotentRing}(R, \cdot)$$

Remark (Negated quantified statement)*.*

$$\neg\, \text{BooleanRing}(R, +, \cdot, -, 0, 1) \iff \neg\, \text{Ring}(R, +, \cdot, -, 0, 1) \;\vee\; \exists p \in R\colon p \cdot p \neq p.$$

Remark (Failure modes)*.* The failure modes are exactly the ways that one of the defining structural conditions can fail.

Remark (Failure mode decomposition)*.* A structure fails to be a Boolean ring in exactly one of two ways:

$$\underbrace{\neg\, \text{Ring}(R, +, \cdot, -, 0, 1)}_{\text{fails a ring axiom}} \qquad \vee \qquad \underbrace{\exists p \in R\colon p \cdot p \neq p}_{\text{some element is not idempotent}}$$

The second failure mode is the structurally interesting one. A ring in which even one element fails $p \cdot p = p$ is not Boolean. The simplest example: the ring $\mathbb{Z}$ of integers fails idempotence at every element except 0 and 1.

Remark (Canonical examples)*.*

- $\mathbf{2} = \{0, 1\}$ with addition mod 2 and ordinary multiplication is the simplest non-degenerate Boolean ring.
- $\mathbf{2}^n$, the set of all n-tuples of elements of $\mathbf{2}$, with coordinatewise addition and multiplication, is a Boolean ring with 2^n elements. The zero is $(0, \ldots, 0)$ and the unit is $(1, \ldots, 1)$.
- $(\mathcal{P}(X), \triangle, \cap)$, the power set of any set X with symmetric difference as addition and intersection as multiplication, is a Boolean ring. Under the correspondence $A \leftrightarrow \mathbf{1}_A$ (the characteristic function of A), this ring is isomorphic to $\mathbf{2}^X$.

Remark (Interpretation). A Boolean ring is the minimal algebraic structure that forces the full Boolean calculus to emerge from a single additional axiom — idempotence of multiplication. The ring axioms supply the additive abelian group and the multiplicative monoid with distributivity. Idempotence then forces characteristic 2 (every element equals its own additive inverse), forces commutativity of multiplication, and identifies the ring with a field of sets via the representation theorem (Halmos–Givant, Chapter 22). The abstract definition thus captures exactly the algebra of set-theoretic operations: symmetric difference plays the role of addition and intersection plays the role of multiplication.

Remark (Dependencies).

- Definition: Ring

Example (The two-element Boolean ring). The simplest non-degenerate Boolean ring is the two-element ring $\mathbf{2} = \{0, 1\}$. Its addition and multiplication are given by the following tables.

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Addition in $\mathbf{2}$ is addition modulo 2: $1 + 1 = 0$. Multiplication in $\mathbf{2}$ is ordinary multiplication of integers restricted to $\{0, 1\}$. Every element of $\mathbf{2}$ satisfies $p + p = 0$ and $p \cdot p = p$, making $\mathbf{2}$ the canonical example against which both axioms (10) and (11) are verified.

Definition — Idempotent Element

Definition 5.3.3 (Idempotent Element)**.** An element p of a ring R is called **idempotent** if

$$p \cdot p = p.$$

A ring in which every element is idempotent is called an **idempotent ring**.

Remark (Standard quantified statement).

$$\text{IdempotentElement}(p, R, \cdot) \iff p \cdot p = p.$$

$$\text{IdempotentRing}(R, \cdot) \iff \forall p \in R\colon p \cdot p = p.$$

Remark (Definition predicate reading).

$\text{IdempotentElement}(p, R, \cdot)$ $\quad$ $\text{IdempotentRing}(R, \cdot) \iff \forall p \in R\colon \text{IdempotentElement}(p, R, \cdot)$

Remark (Negated quantified statement).

$$\neg\ \text{IdempotentElement}(p, R, \cdot) \iff p \cdot p \neq p.$$

Remark (Interpretation). Idempotence is the single additional axiom that forces Boolean structure. In the two-element ring $\mathbf{2}$, verification is immediate: $0 \cdot 0 = 0$ and $1 \cdot 1 = 1$. In a power set algebra $\mathcal{P}(X)$, idempotence of intersection — $A \cap A = A$ — is the set-theoretic counterpart. The force of the axiom is not in these examples but in what it implies for an arbitrary ring: characteristic 2 and commutativity of multiplication both follow from idempotence alone, as the theorems below establish.

Remark (Dependencies).

- Definition: Ring

Definition — Characteristic 2

Definition 5.3.4 (Characteristic 2)**.** A ring R is said to have **characteristic** 2 if

$$p + p = 0 \qquad \text{for all } p \in R. \tag{10}$$

Remark (Standard quantified statement).

$$\text{Characteristic2}(R, +, 0) \iff \forall p \in R\colon p + p = 0.$$

Remark (Definition predicate reading).

$$\text{Characteristic2}(R, +, 0) \iff \forall p \in R\colon \text{IdempotentElement}(p, R, +)$$

Remark (Negated quantified statement).

$$\neg\ \text{Characteristic2}(R, +, 0) \iff \exists p \in R\colon p + p \neq 0.$$

Remark (Interpretation). Characteristic 2 means that every element is its own additive inverse: $p + p = 0$ says precisely $p = -p$. The operation of negation therefore becomes the identity map — applying $-$ to any element returns that element unchanged. Halmos observes that in a ring of characteristic 2, the minus sign for additive inverses is never needed and is henceforth dropped. Axiom (7), $p + (-p) = 0$, is replaced in the official Boolean ring axiom set by the stronger equation (10), $p + p = 0$, which is a theorem derived from idempotence.

Remark (Dependencies).

- Definition: Boolean Ring
- Definition: Idempotent Element

Lemma

Lemma 5.3.5 (Self Additive Inverse)**.** *In any Boolean ring R,*

$$p + p = 0 \qquad \textit{for all } p \in R.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall p \in R\colon p + p = 0.$$

Remark (Definition predicate reading).

$$\text{BooleanRing}(R, +, \cdot, -, 0, 1) \implies \text{Characteristic2}(R, +, 0).$$

Remark (Interpretation). This is Halmos consequence (a): idempotence of multiplication forces the additive structure to have characteristic 2. The proof expands $(p+q)^2$ by distributivity and idempotence, cancels p and q, then sets $p = q$ to obtain $p + p = 0$. The Lean 4 formalisation is `AdditiveIdempotence` in `BooleanAlgebraDefinitions.lean`.

Remark (Dependencies).

- Definition: Boolean Ring
- Definition: Characteristic 2

Lemma

Lemma 5.3.6 (Self Negation). *In any Boolean ring R,*

$$-p = p \qquad \textit{for all } p \in R.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall p \in R\colon\ -p = p.$$

Remark (Interpretation). This follows immediately from 5.3.5. Since $p+p = 0$ and $p+(-p) = 0$ both hold, the uniqueness of additive inverses gives $-p = p$. The negation operation is therefore the identity map on any Boolean ring. Halmos uses this to drop the minus sign entirely from his notation for the rest of the book: since $-p = p$, writing $-p$ and writing p are the same thing.

Remark (Dependencies).

- Lemma: Self Additive Inverse

Remark (No Boolean ring of order 3). A Boolean ring of order 3 does not exist. The additive group $(R, +)$ of any Boolean ring satisfies $p + p = 0$ for all p, by Lemma 5.3.5. Every element therefore has additive order dividing 2, so $(R, +)$ is an elementary abelian 2-group. Elementary abelian 2-groups have order 2^n for some $n \geq 0$. Since 3 is not a power of 2, no Boolean ring of order 3 can exist.

Example (The four-element Boolean ring $\mathbf{2}^2$). The smallest Boolean ring with more than two elements is $\mathbf{2}^2 = \{(0,0),\ (0,1),\ (1,0),\ (1,1)\}$, the set of ordered pairs from $\mathbf{2} = \{0,1\}$. Addition and multiplication are defined coordinatewise:

$$(p_0,p_1) + (q_0,q_1) \;=\; (p_0 + q_0,\ p_1 + q_1), \qquad (p_0,p_1)\cdot(q_0,q_1) \;=\; (p_0\cdot q_0,\ p_1\cdot q_1),$$

where each coordinate is computed in $\mathbf{2}$. The zero element is $(0,0)$ and the unit is $(1,1)$.

Addition table.

$+$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,1)$	$(0,1)$	$(0,0)$	$(1,1)$	$(1,0)$
$(1,0)$	$(1,0)$	$(1,1)$	$(0,0)$	$(0,1)$
$(1,1)$	$(1,1)$	$(1,0)$	$(0,1)$	$(0,0)$

Multiplication table.

$\cdot$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$	$(0,0)$	$(0,0)$	$(0,0)$	$(0,0)$
$(0,1)$	$(0,0)$	$(0,1)$	$(0,0)$	$(0,1)$
$(1,0)$	$(0,0)$	$(0,0)$	$(1,0)$	$(1,0)$
$(1,1)$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$

Verification of idempotence. Every element squares to itself:

p	$p \cdot p$
$(0,0)$	$(0,0)$
$(0,1)$	$(0,1)$
$(1,0)$	$(1,0)$
$(1,1)$	$(1,1)$

Verification of characteristic 2**.** Every element is its own additive inverse:

p	$p + p$
$(0,0)$	$(0,0)$
$(0,1)$	$(0,0)$
$(1,0)$	$(0,0)$
$(1,1)$	$(0,0)$

Remark (Boolean rings have order 2^n). Every finite Boolean ring is isomorphic to $\mathbf{2}^n$ for some $n \geq 0$, and therefore has exactly 2^n elements. This follows from the representation theorem (Halmos–Givant, Chapter 22): every Boolean algebra is isomorphic to a field of sets, and every finite field of sets is isomorphic to $\mathcal{P}(X)$ for some finite set X with $|X| = n$, giving $|\mathcal{P}(X)| = 2^n$. The permissible finite sizes are $1, 2, 4, 8, 16, \ldots$ and no other cardinality is achievable.

5.4 Sigma Algebras

5.5 Borel Sets

Proofs

Remark (Return). Return to Lemma

Remark (Proof vault). Handwritten proof record

Lemma (Self Additive Inverse). *In any Boolean ring R,*

$$p + p = 0 \qquad \text{for all } p \in R.$$

Professional Standard Proof.
Apply idempotence to the element $p + p$:

$$(p + p) \cdot (p + p) = p + p. \tag{i}$$

Expand the left side using right distributivity (9), then left distributivity (8), then collapse each $p \cdot p = p$ by idempotence (11):

$$(p+p)\cdot(p+p) \overset{(9)}{=} p\cdot(p+p)+p\cdot(p+p) \overset{(8)}{=} (p^2+p^2)+(p^2+p^2) \overset{p^2=p}{=} (p+p)+(p+p). \tag{ii}$$

Combining (i) and (ii): $(p + p) + (p + p) = p + p$. Setting $s = p + p$, the equation $s + s = s$ gives $s = 0$ by adding $-s$ and applying associativity, the inverse law, and the identity law. Hence $p + p = 0$. □

Detailed Learning Proof.

Step 1. Apply idempotence to the element $p + p$. The element $p + p$ is a member of the carrier of R. Axiom (11) (multiplicative idempotence) asserts every element of R is its own square. Apply this to the element $p + p$:

$$(p + p) \cdot (p + p) = p + p. \tag{i}$$

This is the pivot of the proof. It says $p + p$ squares to itself. Nothing about $p + p = 0$ has been assumed; that is the conclusion to be derived.

Step 2. Expand $(p + p) \cdot (p + p)$ using right distributivity. Axiom (9) (right distributivity) states:

$$(q + r) \cdot t = q \cdot t + r \cdot t \qquad \forall\, q, r, t \in R.$$

Set $q = p$, $r = p$, $t = p + p$:

$$(p + p) \cdot (p + p) = p \cdot (p + p) \;+\; p \cdot (p + p). \tag{ii}$$

The product has split into two identical copies of $p \cdot (p + p)$. This step holds in any ring; no idempotence has been used.

Step 3. Expand each $p \cdot (p + p)$ using left distributivity. Axiom (8) (left distributivity) states:

$$t \cdot (q + r) = t \cdot q + t \cdot r \qquad \forall\, t, q, r \in R.$$

Apply to $p \cdot (p + p)$ with $t = p$, $q = p$, $r = p$:

$$p \cdot (p + p) = p \cdot p \;+\; p \cdot p.$$

Apply to both copies in (ii):

$$p \cdot (p + p) + p \cdot (p + p) = (p \cdot p + p \cdot p) + (p \cdot p + p \cdot p). \tag{iii}$$

The right side contains four occurrences of $p \cdot p$. This expansion holds in any ring; idempotence has still not been used.

Step 4. Collapse each $p \cdot p$ to p by idempotence. Axiom (11) gives $p \cdot p = p$. Substitute into all four occurrences in (iii):

$$(p \cdot p + p \cdot p) + (p \cdot p + p \cdot p) = (p + p) + (p + p). \tag{iv}$$

This is the second and final use of idempotence.

Step 5. Chain (i) through (iv). Equations (i), (ii), (iii), (iv) combine as:

$$p + p \;=\; (p + p) \cdot (p + p) \;=\; (p + p) + (p + p).$$

Hence

$$(p + p) + (p + p) = p + p. \tag{v}$$

Step 6. Conclude $p + p = 0$ by cancellation. Let $s = p + p$. Equation (v) states $s + s = s$. Add the additive inverse $-s$ to the right of both sides, then apply associativity, the inverse law, and the identity law:

$$s{+}s = s \implies (s{+}s){+}(-s) = s{+}(-s) \implies s{+}\underbrace{(s + (-s))}_{=0} = 0 \implies s{+}0 = 0 \implies s = 0.$$

Since $s = p + p$, the conclusion is $p + p = 0$. $\square$

Remark (Proof structure). The proof is direct. The strategy applies idempotence twice: once to $p + p$ as a whole (Step 1) and once to p individually (Step 4), with distributivity (Steps 2–3) connecting the two applications. The conclusion follows by a standard additive cancellation (Step 6) using only the group axioms for $(R, +)$. No circularity arises: the proof never assumes $1 + 1 = 0$ or any consequence of characteristic 2. Both $p + p = 0$ and $1 + 1 = 0$ are consequences of idempotence; this proof derives the former directly without appealing to the latter. The Lean 4 formalisation is `AdditiveIdempotence` in `LRA/VolumeI/BooleanAlgebra/BooleanAlgebraDefinitions.lean`.

Remark (Dependencies).

- Definition of Ring — axioms (1) associativity, (5) identity, (7) inverse, (8) left distributivity, (9) right distributivity.
- Definition of Boolean Ring — axiom (11) multiplicative idempotence, applied in Steps 1 and 4.
- Definition of Idempotent Element — names the property $p \cdot p = p$.

Remark (Return). Return to Lemma

Remark (Proof vault). Handwritten proof record

Lemma (Self Negation)**.** *In any Boolean ring R,*

$$-p = p \qquad \textit{for all } p \in R.$$

Professional Standard Proof.
By Self Additive Inverse, $p + p = 0$. By the additive inverse axiom (7), $p + (-p) = 0$. Since both p and $-p$ satisfy the equation $p + x = 0$, and additive inverses are unique in any group (a standard group theory fact applied to the abelian group $(R, +)$), we conclude $-p = p$. □

Detailed Learning Proof.
Step 1. Recall the two equations that both hold in R.
From Self Additive Inverse (Lemma 5.3.5):

$$p + p = 0. \tag{i}$$

From the additive inverse axiom (7) of the ring definition:

$$p + (-p) = 0. \tag{ii}$$

Step 2. Apply uniqueness of additive inverses.
The additive group $(R, +, 0)$ is a group (abelian, by axiom (3)). In any group, if $a + b = 0$ and $a + c = 0$, then:

$$b = 0 + b = (a + c) + b \overset{\text{assoc}}{=} a + (c + b) \overset{\text{comm}}{=} a + (b + c) \overset{\text{assoc}}{=} (a + b) + c = 0 + c = c.$$

Apply with $a = p$, $b = p$ (from (i)) and $c = -p$ (from (ii)): both p and $-p$ are additive inverses of p, so $p = -p$. □

Remark (Proof structure). A one-step uniqueness argument. Self Additive Inverse shows p is its own additive inverse; the ring axiom asserts $-p$ is also an additive inverse of p; uniqueness of inverses in the underlying group forces $p = -p$. The proof has no calculation content — it is purely structural.

Remark (Dependencies).

- Lemma: Self Additive Inverse
- Definition: Ring (axiom (7), additive inverse law)

Chapter 6

Measure Spaces

Proofs

Bibliography

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